Chapter 1

Questions on Lesson Four

Kirchhoff's Laws

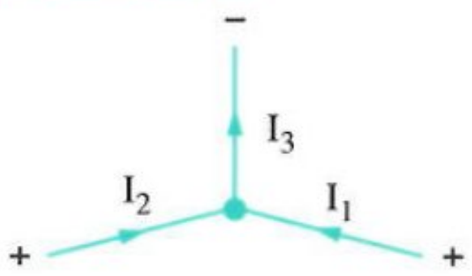
Interactive test

The questions signed by ✱ are answered in detail. ● Understand ○ Apply ⁘ Higher Order Thinking Skills

Important guidelines

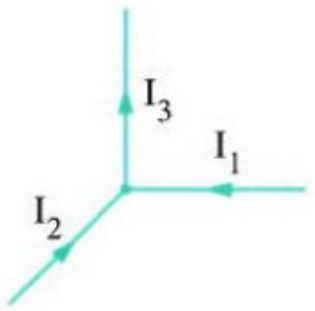
⇒ **Kirchhoff's laws:**

• **Kirchhoff's first law :**



$\Sigma I = 0$

$I_1 + I_2 - I_3 = 0$



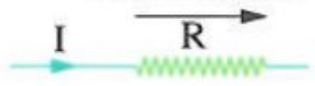
$\Sigma I_{in} = \Sigma I_{out}$

$I_1 + I_2 = I_3$

• **Kirchhoff's second law :**

You must consider the following sign rules when applying Kirchhoff's second law in a closed loop:

(1) When using the mathematical formula: $\Sigma V_B = \Sigma IR$

In the resistor	In the battery
Path direction →  $V = +IR$	Path direction ←  $V = V_B$

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Path direction I R $V = -IR$	Path direction + − $V = -V_B$

(2) When using the mathematical formula: $\Sigma V = 0$

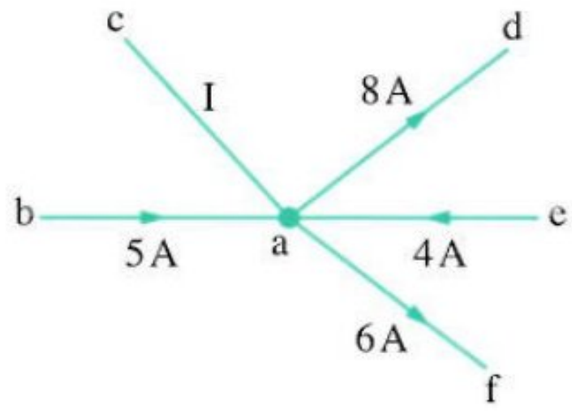
In the resistor	In the battery
Path direction + R − $V = -IR$	Path direction + − $V = -V_B$
Path direction + R − $V = IR$	Path direction + − $V = V_B$

First Multiple choice questions

Kirchhoff's first law

1 In the network shown in the figure, ...

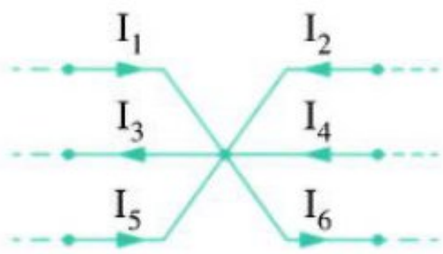
	Electric current (I)	Current direction (I)
a	3 A	from a to c
b	3 A	from c to a
c	5 A	from a to c
d	5 A	from c to a



2 In the opposite figure, if $I_1 = I_2 = I_3 = I_4 = I_5$, then I_6 = ...

a I_1 b $2\,I_1$

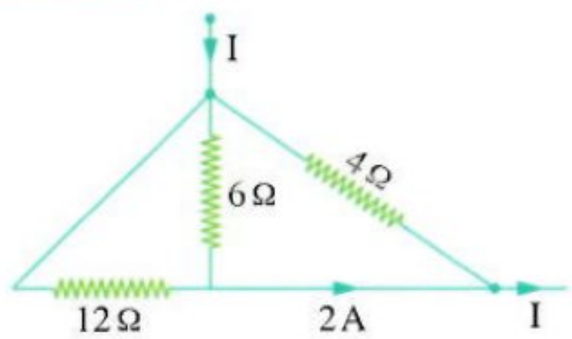
c $3\,I_1$ d $4\,I_1$



3 In the opposite figure, the value of I is ..... .

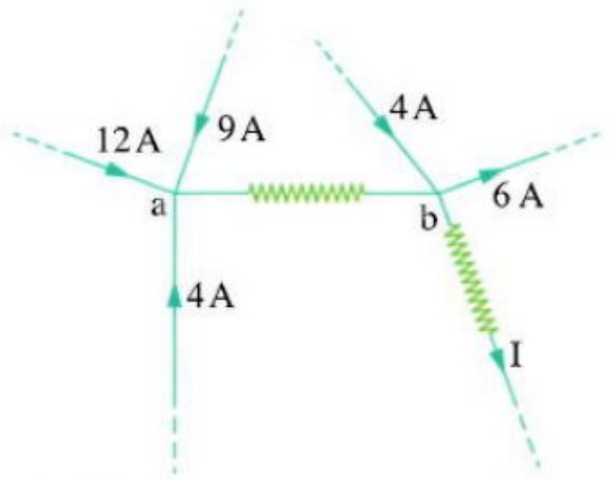
a 2 A b 4 A

c 6 A d 12 A



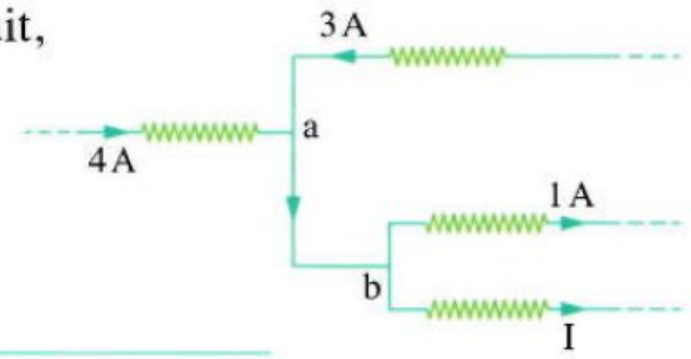
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4 The opposite figure represents a part of an electric circuit, hence the value of I is ..... .



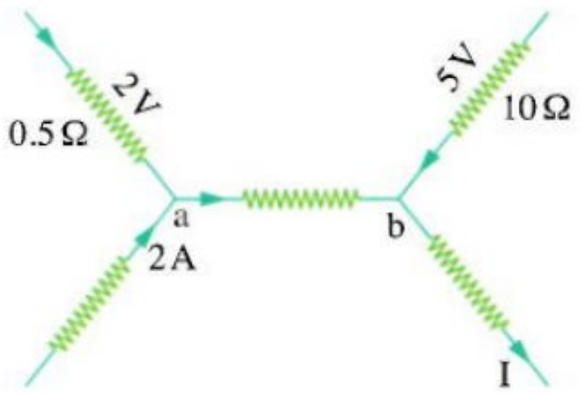
(a) 19 A
(b) 23 A
(c) 27 A
(d) 31 A

5 The opposite figure represents a part of an electric circuit, hence the value of I is ..... .



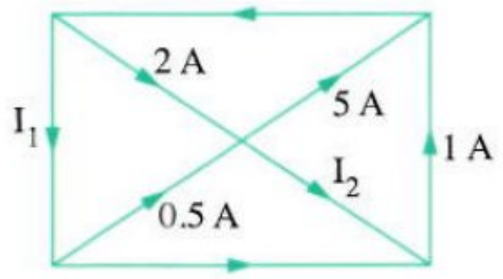
(a) 2 A (b) 4 A
(c) 6 A (d) 8 A

6 The opposite figure represents a part of an electric circuit, hence the value of I is ..... .



(a) 2.5 A
(b) 3.5 A
(c) 6.5 A
(d) 7.5 A

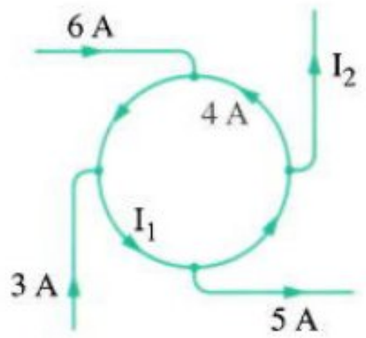
7 In the opposite circuit:



(i) The intensity of current I_1 is ..... .
(a) – 2.5 A (b) 3.5 A (c) 4 A (d) 6 A

(ii) The intensity of current I_2 is ..... .
(a) – 2.5 A (b) 3.5 A (c) 4 A (d) 6 A

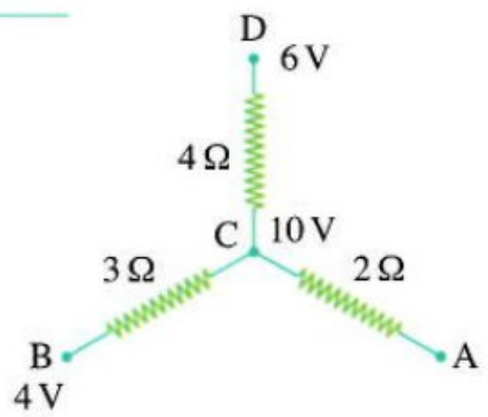
8 A metal ring is embedded in an electric circuit through which electric currents pass as shown in the opposite figure. So:



(i) I_1 equals ..... .
(a) 0 A (b) 4 A (c) 9 A (d) 13 A

(ii) I_2 equals ..... .
(a) 2 A (b) 4 A (c) 9 A (d) 12 A

9 In the opposite electric circuit, if the electric potentials of the points B, C and D are 4 V, 10 V and 6 V respectively, then the electric potential of point A equals ..... .

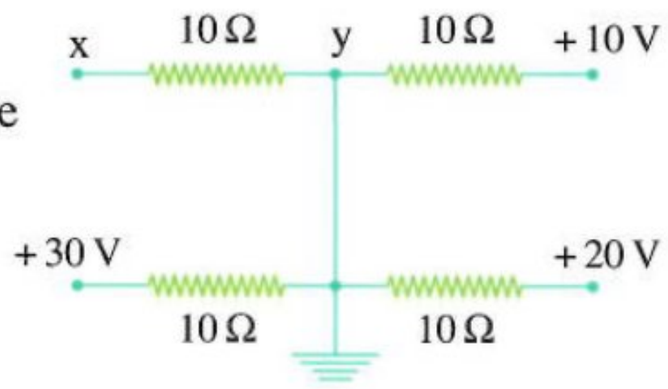


(a) 4 V (b) 8 V
(c) 12 V (d) 16 V

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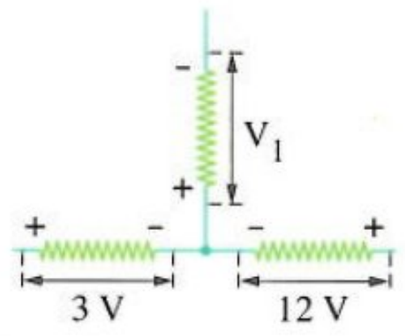
10 The opposite figure shows a part of an electrical circuit indicating the electric potentials at some points. Then, the intensity of the electric current passing between the two points (x) and (y) shown in the figure equals ..... .



(a) 1 A (b) 4 A

(c) 5 A (d) 6 A

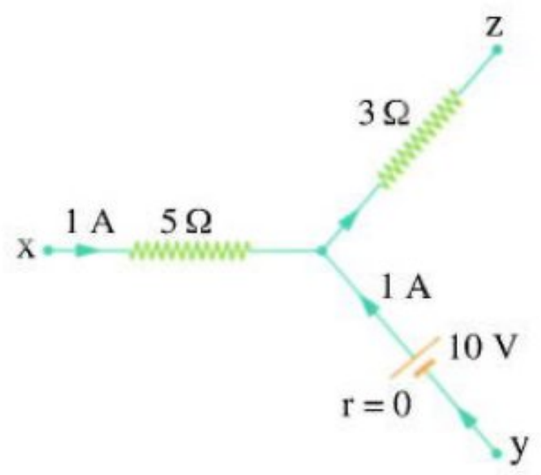
11 The opposite diagram represents a part of a closed electric circuit, so if each resistor has a resistance of 3 Ω, the value of V_1 is ..... .



(a) 10 V (b) 20 V

(c) 15 V (d) 12 V

12 In the opposite figure, the correct arrangement for the electric potential at points x, y and z is .....



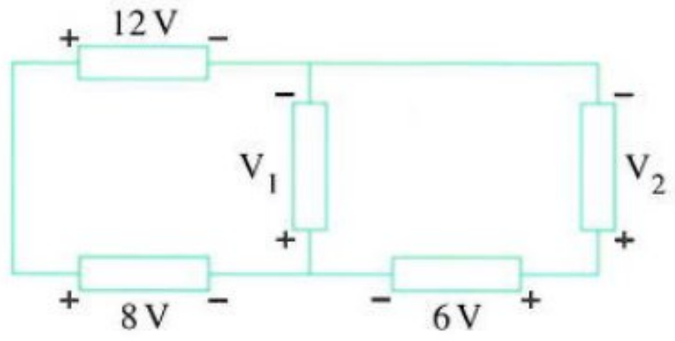
(a) $V_x > V_y > V_z$ (b) $V_y > V_x > V_z$

(c) $V_z > V_x > V_y$ (d) $V_x > V_z > V_y$

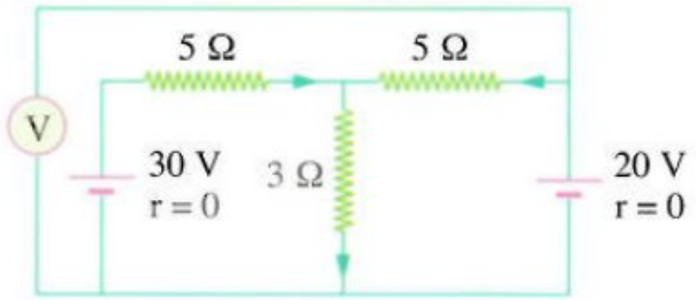
Kirchhoff's second law

13 In the shown circuit, the values of ..... .

	V_1	V_2
(a)	4 V	10 V
(b)	4 V	7 V
(c)	10 V	10 V
(d)	10 V	7 V



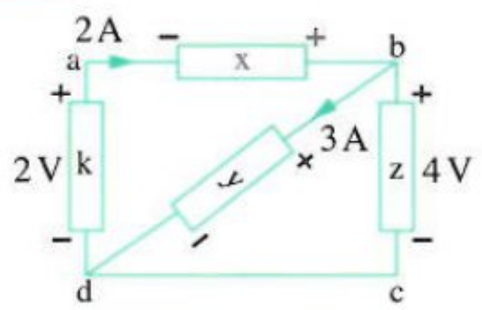
14 In the opposite electric circuit, the reading of the voltmeter is ..... .



(a) 20 V (b) 15 V

(c) 5 V (d) 0

15 * In the opposite circuit, the powers $(P_w)_x$ and $(P_w)_y$ are ..... respectively.

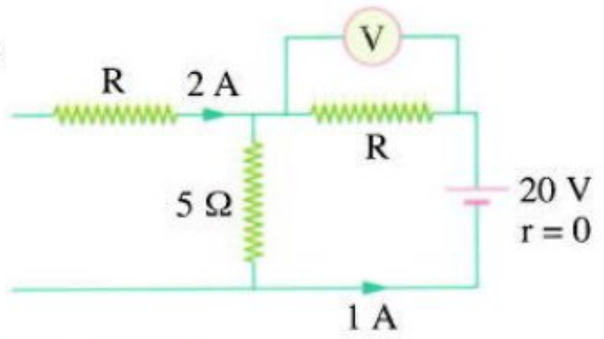


(a) 4 W, 12 W (b) 12 W, 4 W

(c) 16 W, 12 W (d) 4 W, 14 W

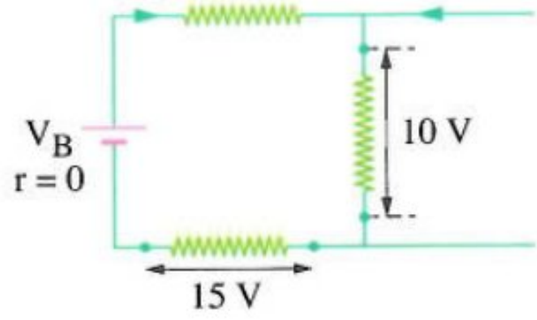
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16 The opposite diagram represents a part of an electric circuit, so the voltmeter reading is ..... .



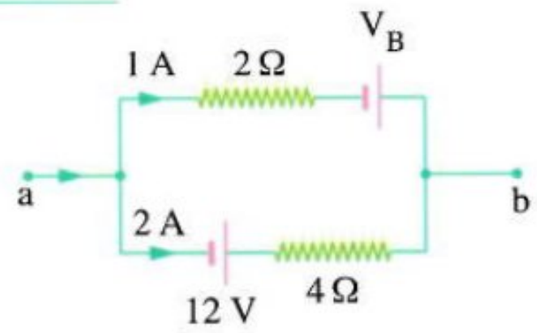
(a) 20 V (b) 15 V
(c) 5 V (d) 1 V

17 The opposite figure represents a part of an electric circuit through which an electric current is passing, so the value of V_B is ..... .



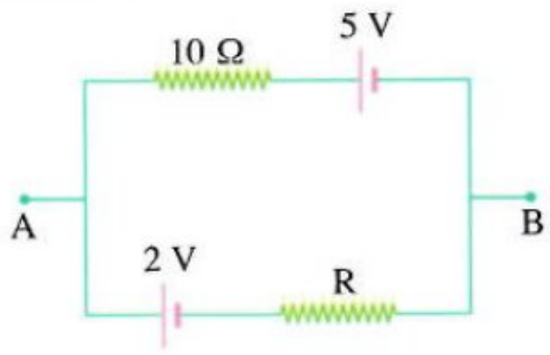
(a) 20 V (b) 30 V
(c) 40 V (d) 50 V

18 The opposite figure represents a part of an electric circuit, so the magnitude of the electromotive force V_B equals ..... .



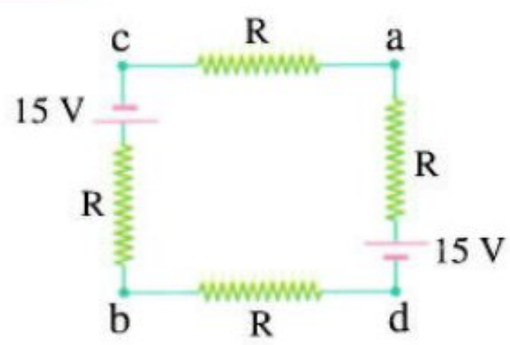
(a) 3 V (b) 4 V
(c) 6 V (d) 8 V

19 If the potential difference between A and B in the opposite electric circuit equals 4 V, then the value of resistance R equals ..... .



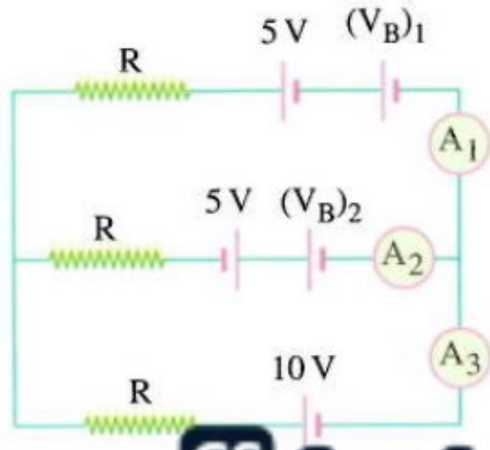
(a) 5 Ω (b) 10 Ω
(c) 15 Ω (d) 20 Ω

20 * In the electric circuit shown in the figure, if the value of R is 7.5 Ω, the potential difference between points a and b equals ..... .



(a) 0 (b) 5 V
(c) 10 V (d) 15 V

21 In the opposite electrical circuit, if the reading of each of the three ammeters is equal to zero, and all the batteries have negligible internal resistances, then the values of the electromotive forces $(V_B)_1$ and $(V_B)_2$ respectively are ..... .

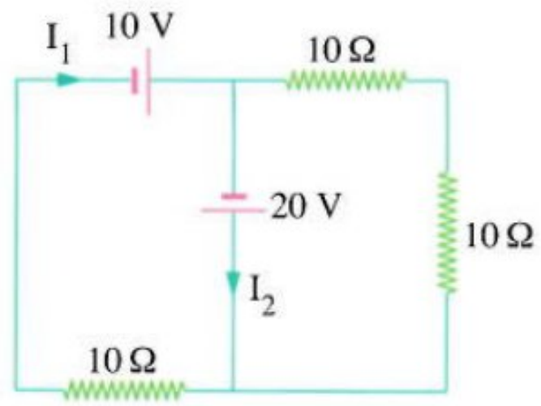


(a) 5 V, 10 V
(b) 10 V, 5 V
(c) 5 V, 15 V
(d) 15 V. 5 V

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22 In the opposite electric circuit:



(i) The current intensity I_1 equals ..... .

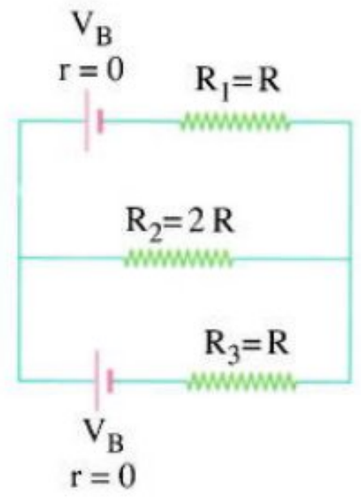
(a) 1 A (b) 2 A

(c) 3 A (d) 4 A

(ii) The current intensity I_2 equals ..... .

(a) 1 A (b) 2 A (c) 3 A (d) 4 A

23 The opposite figure shows a closed electric circuit, so the resistance which carries the greatest electric current intensity is ..... .



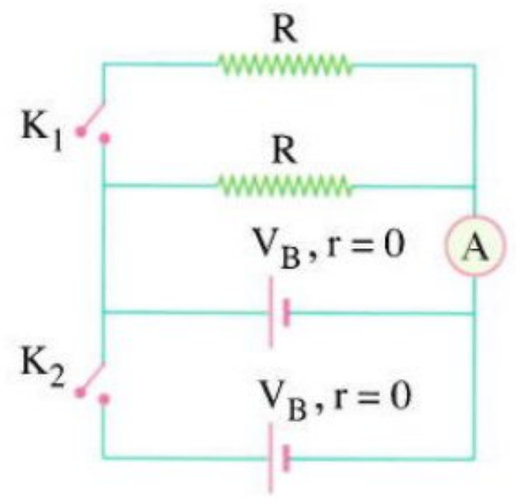
(a) R_1

(b) R_2

(c) R_3

(d) R_1, R_2

24 In the opposite electrical circuit, if the ammeter reading is I, so its reading when:



(i) closing switch (K_1) only will be ..... .

(a) I (b) $\frac{I}{2}$

(c) 2 I (d) 4 I

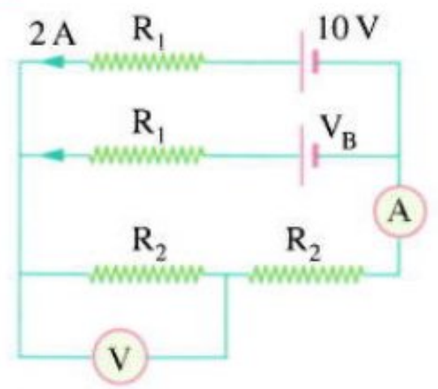
(ii) closing switch (K_2) only will be ..... .

(a) I (b) $\frac{I}{2}$ (c) 2 I (d) 4 I

(iii) closing both switches (K_1 and K_2) together will be ..... .

(a) I (b) $\frac{I}{2}$ (c) 2 I (d) 4 I

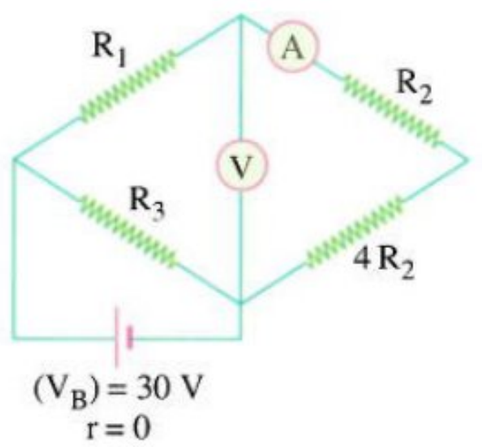
25 ✱ In the opposite circuit, if the ammeter reading is 4 A and the voltmeter reading is 3 V, the electromotive force (V_B) equals ..... .



(a) 4 V (b) 6 V

(c) 10 V (d) 14 V

26 In the opposite electrical circuit, if the ammeter reading is 1 A and the voltmeter reading is 10 V, then the values of the resistances (R_1) and (R_2) respectively are ....., ...

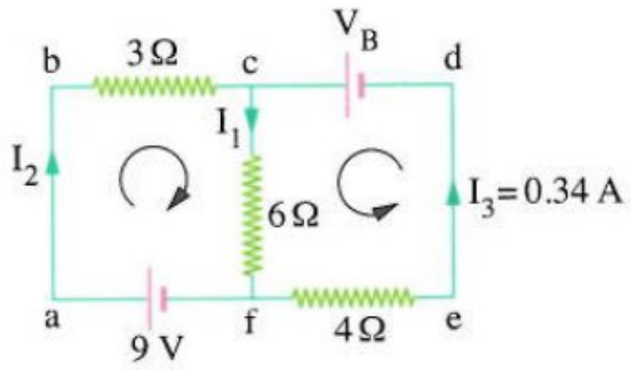


(a) 10 Ω, 5 Ω (b) 30 Ω, 2 Ω

(c) 20 Ω, 10 Ω (d) 20 Ω, 2 Ω

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27 ✳ In the opposite electric circuit:



(i) The intensity of current I_1 equals ..... .

(a) 4.01 A (b) 8.02 A

(c) 0.77 A (d) 1.11 A

(ii) The intensity of current I_2 equals ..... .

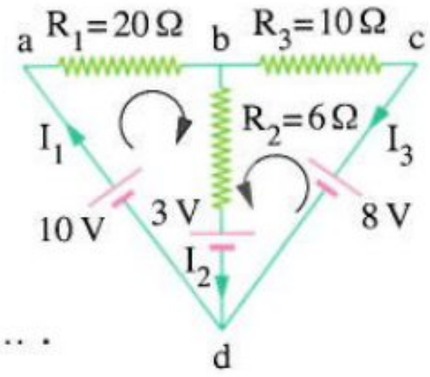
(a) 0.77 A (b) 4.01 A (c) 8.02 A (d) 1.11 A

(iii) The electromotive force V_B equals ..... .

(a) 2.01 V (b) 4.03 V (c) 6.01 V (d) 8.02 V

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28 ✳ In the opposite electric circuit:



(i) The intensity of the current that passes in resistor R_1 equals ..... .

(a) 0.22 A (b) – 0.23 A

(c) 0.45 A (d) 0.53 A

(ii) The intensity of the current that passes in resistor R_2 equals ..... .

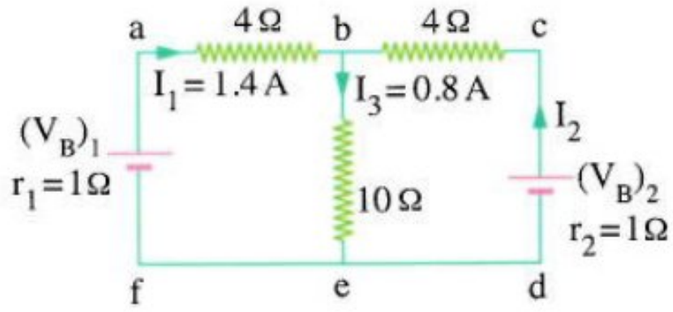
(a) 0.22 A (b) – 0.23 A (c) 0.45 A (d) 0.53 A

(iii) The intensity of the current that passes in resistor R_3 equals ..... .

(a) 0.22 A (b) – 0.23 A (c) 0.45 A (d) 0.53 A

---

29 ✳ In the opposite electric circuit:



(i) The electromotive force $(V_B)_1$ equals ..... .

(a) 5 V (b) 8 V

(c) 15 V (d) 20 V

(ii) The electromotive force $(V_B)_2$ equals ..... .

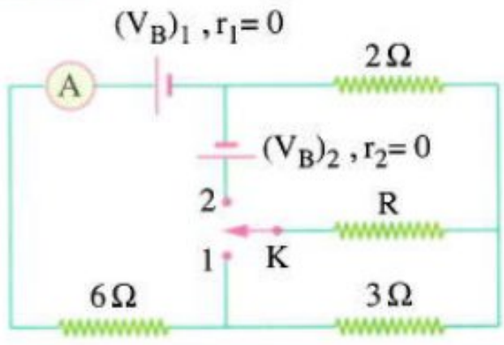
(a) 5 V (b) 8 V (c) 10 V (d) 12 V

(iii) The magnitude of potential difference between points e and b equals ..... .

(a) 5 V (b) 6 V (c) 7 V (d) 8 V

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30 In the opposite electric circuit, the ammeter reading is 2 A, so if when closing switch K in position (1), the ammeter reading becomes 2.2 A and when closing switch K in position (2), the ammeter reading becomes 1.81 A, then:



(i) The electromotive force of battery $(V_B)_1$ is ..... .

(a) 20 V (b) 22 V (c) 18 V (d) 16 V

(ii) The resistance of R equals ..... .

(a) 3 Ω (b) 6 Ω (c) 2 Ω (d) 12 Ω

(iii) The electromotive force of battery $(V_B)_2$ is approximately ..... .

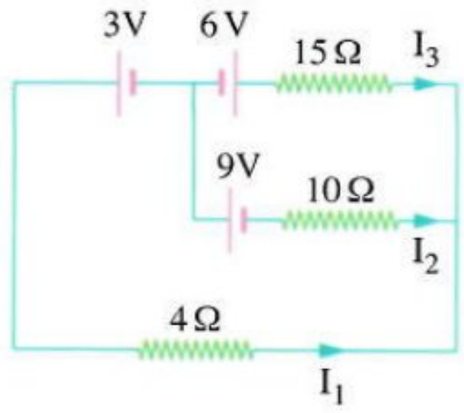
(a) 12 V (b) 14 V (c) 16 V (d) 18 V

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31 ✳ By applying Kirchhoff's laws on the opposite circuit, you can find that:



(Knowing that: The directions indicated on the figure are hypothetical directions and are not necessarily the correct directions for the current)

(i) The intensity of electric current I_1 is ..... .

(a) 0.36 A (b) 0.6 A (c) – 0.36 A (d) – 0.6 A

(ii) The intensity of electric current I_2 is ..... .

(a) 0.96 A (b) 0.6 A (c) – 0.96 A (d) – 0.6 A

(iii) The intensity of electric current I_3 is ..... .

(a) 0.36 A (b) 0.96 A (c) – 0.36 A (d) – 0.96 A

32 A student connected 6 V and 12 V batteries of negligible internal resistances with two identical ohmic resistors, two ammeters and a switch to form an electric circuit as shown in figure (1), then he reversed the terminals of the 6 V battery so that the circuit became as in figure (2):

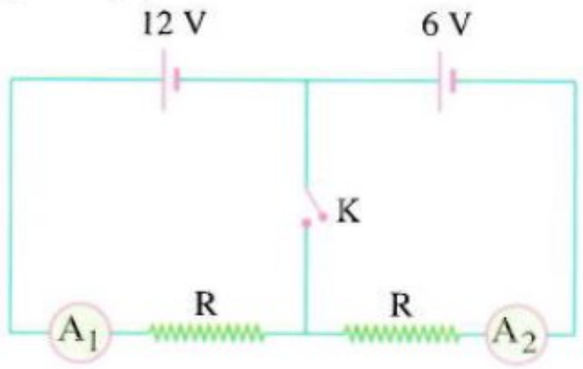


Figure (1)

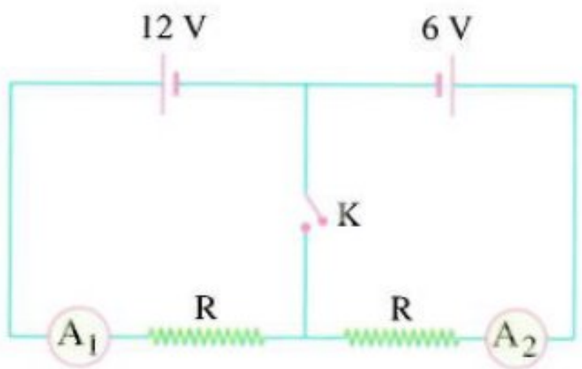


Figure (2)

When the switch (K) is closed, what happens to the readings of the two ammeters A_1 and A_2 respectively in:

(i) figure (1)?

(a) Decreases, increases. (b) Decreases, doesn't change.

(c) Doesn't change, decreases. (d) Increases, decreases.

(ii) figure (2)?

(a) Increases, doesn't change. (b) Increases, increases.

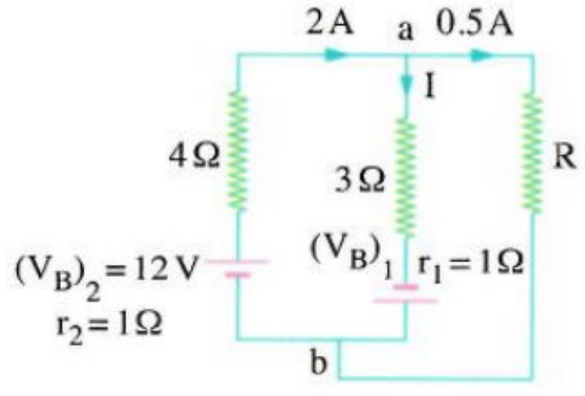
(c) Decreases, decreases. (d) Doesn't change, doesn't change.

33 ✳ From the opposite circuit diagram:

(i) The magnitude of potential difference between the two points a, b equals ..... .

(a) 1 V (b) 2 V

(c) 4 V (d) 6 V



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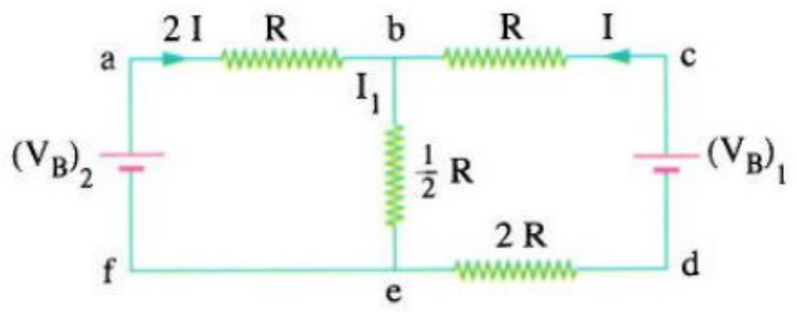
(ii) The electromotive force $(V_B)_1$ equals ..... .

(a) 9 V (b) 6 V (c) 4 V (d) 2 V

(iii) The resistance R is ..... .

(a) 2.5 Ω (b) 2 Ω (c) 4 Ω (d) 7.5 Ω

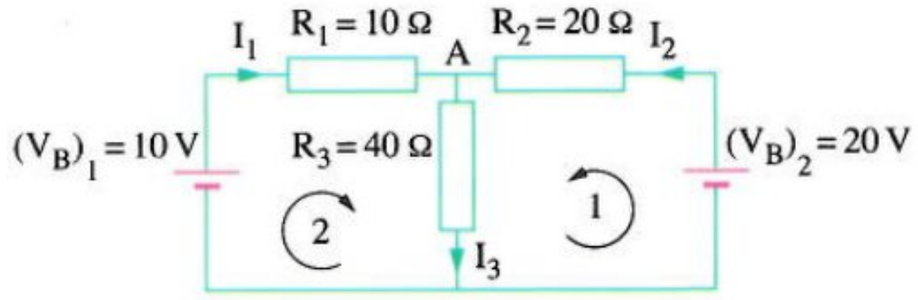
34 ✳ From the opposite circuit, the ratio $\left(\frac{(V_B)_1}{(V_B)_2}\right)$ equals ..... .



(a) $\frac{4}{5}$ (b) $\frac{5}{4}$ (c) $\frac{9}{7}$ (d) $\frac{7}{9}$

35 ✳ In the opposite electric circuit:

(i) The intensity of the electric current that passes in resistor R_3 is ..... .

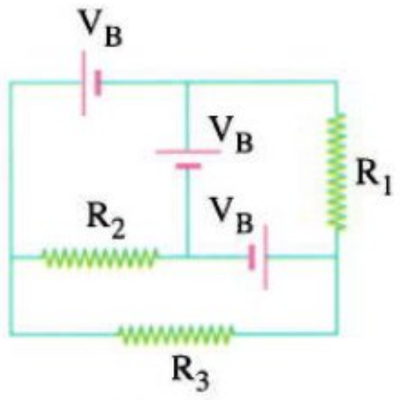


(a) $\frac{5}{3}$ A (b) $\frac{4}{7}$ A (c) $\frac{2}{7}$ A (d) $\frac{4}{9}$ A

(ii) The consumed power in the circuit is ..... .

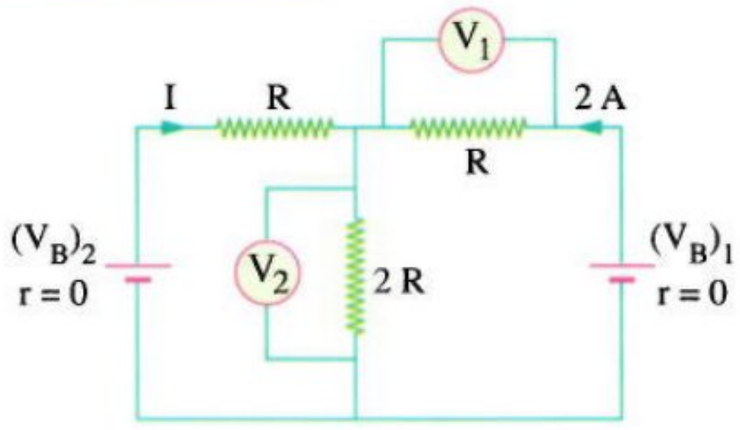
(a) 8.57 W (b) 4.42 W (c) 3.51 W (d) 2.31 W

36 The opposite circuit contains multiple identical electric cells of negligible internal resistances, so which resistance doesn't carry any electric current?



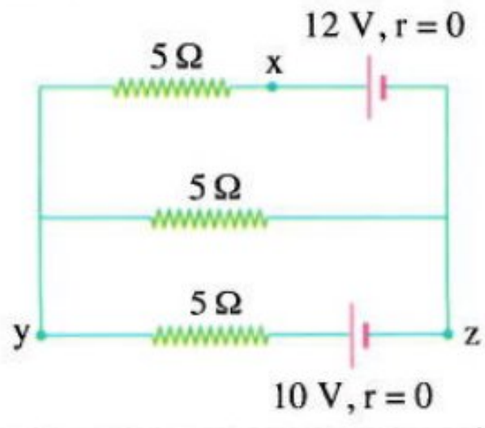
(a) R_1 (b) R_2 (c) R_3 (d) R_1, R_2

37 ✳ The opposite figure represents an electric circuit. If $V_2 = 4\ V_1$, then the value of I equals ..... .



(a) 2 A
(b) 4 A
(c) 6 A
(d) 8 A

38 The opposite diagram represents a closed electric circuit, so .....



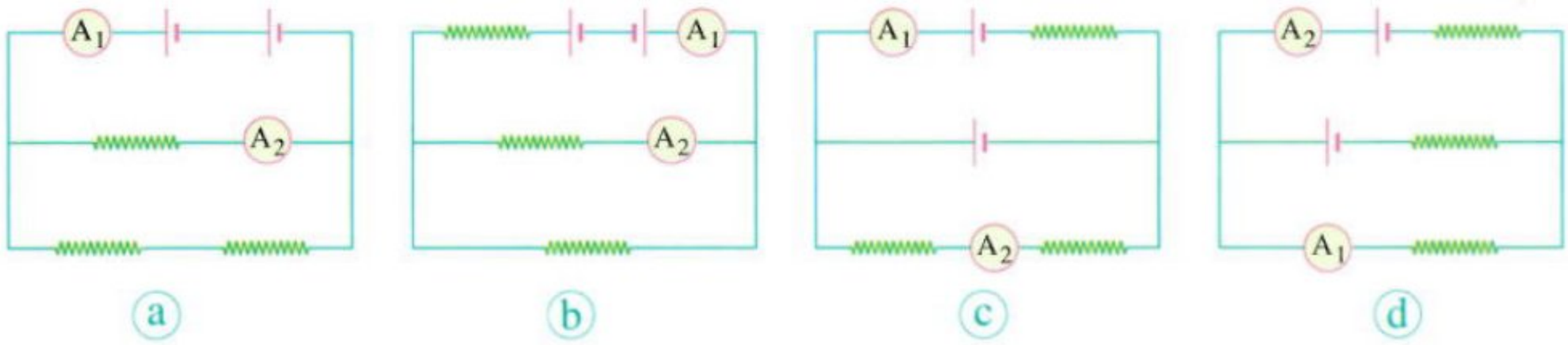
(a) $V_{xy} > V_{xz}$ (b) $V_{xy} = V_{xz}$ (c) $V_{xy} < V_{xz}$ (d) $V_{xy} = V_{yz}$

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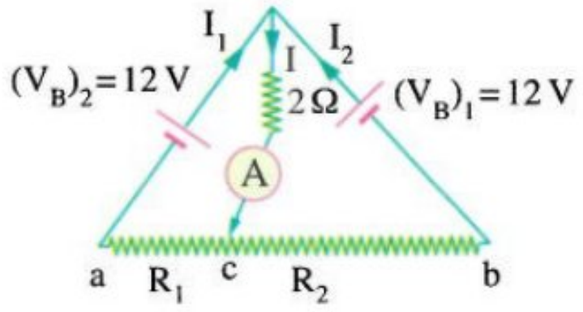
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39 In the following electric circuits, all resistors are identical and all batteries are identical and have negligible internal resistances, so in which of these circuits is the ammeter reading A_1 double of the ammeter reading A_2?



40 ✱ In the opposite electric circuit, if wire ab has a homogeneous resistance of 10 Ω and point c is the point of contact between the slider and wire ab which divides the wire into two resistors R_1 and R_2 where $R_2 = 1.5\ R_1$, so:



(i) The intensity of current I_1 equals ..... .

(a) 0.93 A (b) 1.09 A (c) 1.64 A (d) 2.73 A

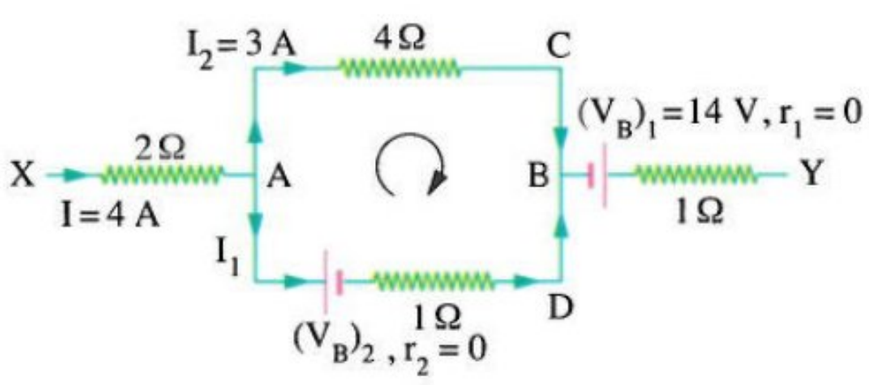
(ii) The intensity of current I_2 equals ..... .

(a) 0.93 A (b) 1.09 A (c) 1.64 A (d) 2.73 A

(iii) The intensity of current I equals ..... .

(a) 0.93 A (b) 1.09 A (c) 1.64 A (d) 2.73 A

41 ✱ The opposite figure represents a part of an electric circuit, using Kirchhoff's laws and considering the shown data and the directions of currents and loops on the figure, so:



(i) The potential difference between the points X and Y equals ..... .

(a) 5 V (b) 7 V

(c) 10 V (d) 11 V

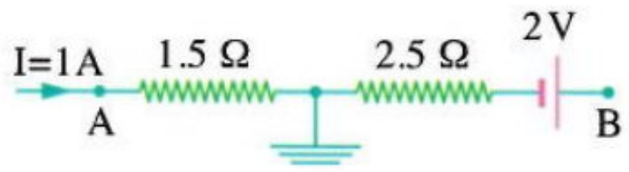
(ii) The magnitude of emf of battery $(V_B)_2$ equals ..... .

(a) 4 V (b) 8 V

(c) 9.5 V (d) 11 V

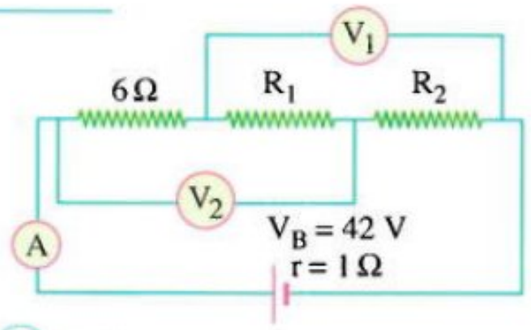
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42 The opposite figure shows a part of an electric circuit through which a current of intensity 1 A passes, so ..... .



	The potential at point A (V_A)	The potential at point B (V_B)
a	– 1.5 V	2 V
b	1.5 V	2 V
c	– 1.5 V	0.5 V
d	1.5 V	– 0.5 V

43 ✱ In the opposite electrical circuit, if the voltmeter readings V_1 and V_2 are 21 V and 24 V respectively, then:



(i) The ammeter reading is ..... .

(a) 1 A (b) 2 A (c) 3 A (d) 4 A

(ii) The value of resistance R_1 equals ..... .

(a) 2 Ω (b) 4 Ω (c) 5 Ω (d) 6 Ω

(iii) The value of resistance R_2 equals ..... .

(a) 2 Ω (b) 4 Ω (c) 5 Ω (d) 6 Ω

44 ✱ The following figure shows a part of an electric circuit, if the consumed power between the two points a and b equals 210 W and the internal resistances of the electric cells are negligible, so:



(i) The unknown electromotive force (V_B) equals ..... .

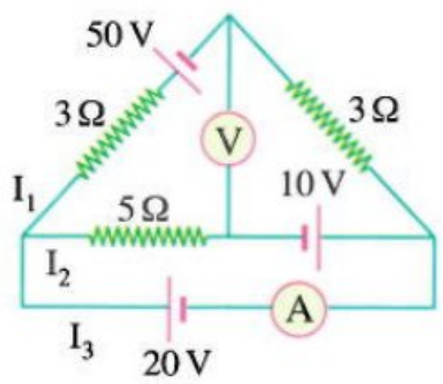
(a) 30 V (b) 20 V (c) 10 V (d) 5 V

(ii) The potential difference between the two points a and b equals ..... .

(a) 20 V (b) 30 V (c) 40 V (d) 50 V

45 ✱ In the opposite electric circuit, the readings of the ammeter and voltmeter are ..... , ...

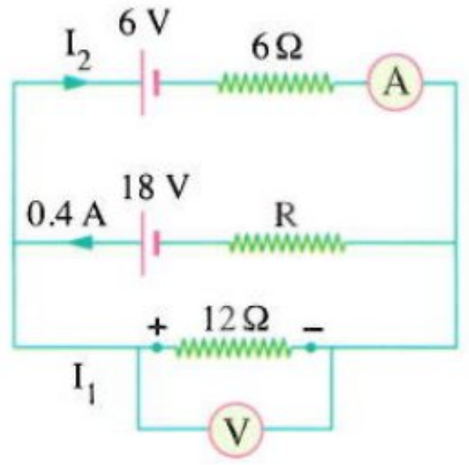
	The ammeter's reading	The voltmeter's reading
a	2 A	5 V
b	1 A	5 V
c	2 A	3 V
d	1 A	3 V



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46 In the opposite electric circuit:



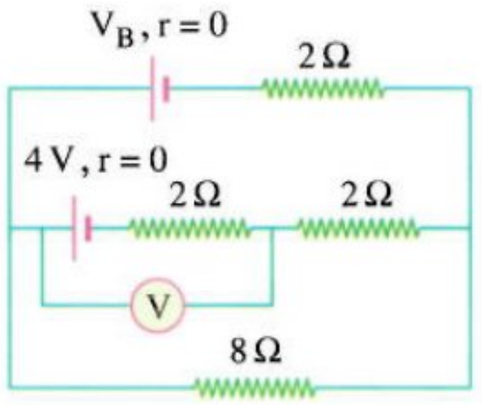
(i) The readings of the ammeter and voltmeter are ..... , ...

	The ammeter's reading	The voltmeter's reading
a	0.067 A	5.6 V
b	0.067 A	3.7 V
c	4.067 A	5.6 V
d	4.067 A	3.7 V

(ii) The resistance of R is ..... .

a 11 Ω b 12 Ω c 31 Ω d 36 Ω

47 In the opposite electrical circuit, if the voltmeter reading is 6 V, then the electromotive force (V_B) of the battery equals ..... .



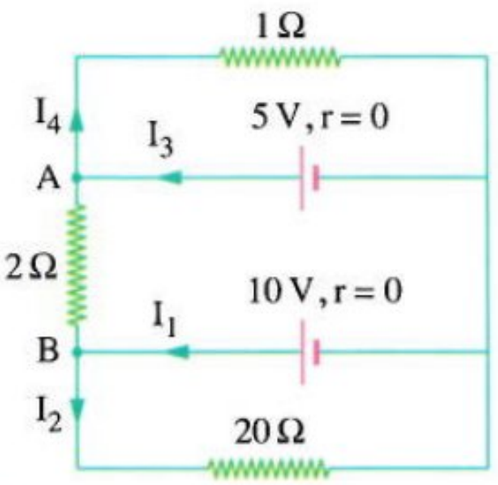
a 4 V

b 8 V

c 10 V

d 12 V

48 In the opposite electric circuit:



(i) The current intensity and its direction through the 2 Ω resistor are ..... .

a 1.5 A, from A to B b 1.5 A, from B to A

c 2.5 A, from A to B d 2.5 A, from B to A

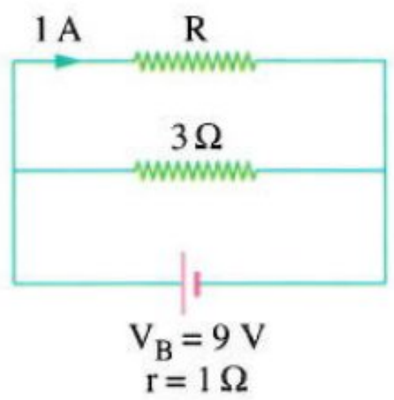
(ii) The current intensity I_1 equals ..... .

a 1 A b 2 A c 3 A d 4 A

(iii) The current intensity I_3 equals ..... .

a 1.5 A b 2.5 A c 3.5 A d 4.5 A

49 In the opposite electrical circuit, if the intensity of the electric current passing through resistor R is 1 A, then the value of resistor R equals ..... .

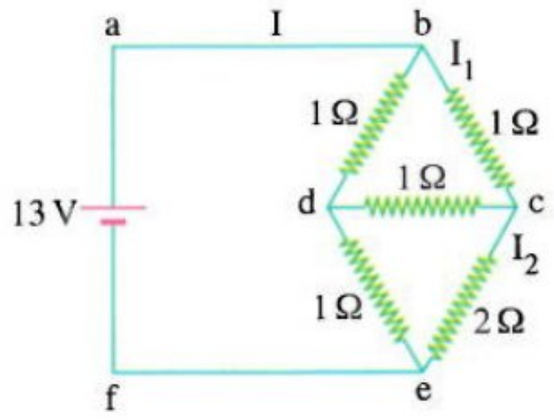


a 1.5 Ω b 3 Ω

c 6 Ω d 12 Ω

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50 ✳ In the opposite electric circuit, the total resistance of the circuit equals ..... .



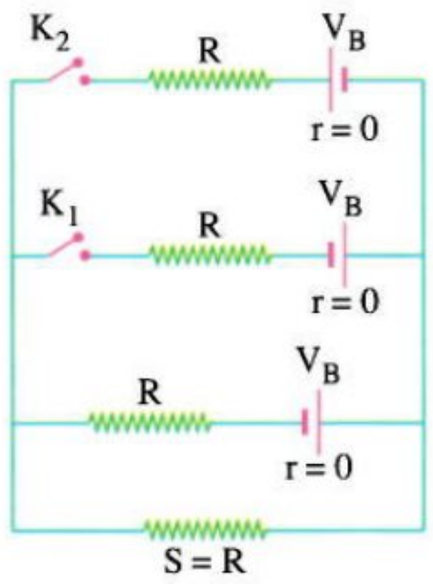
(a) 0.5 Ω

(b) 1 Ω

(c) 1.18 Ω

(d) 1.25 Ω

51 In the opposite electric circuit, what happens to the electric power consumed in resistor S when closing:



(i) switch K_1 only?

(a) Increases.

(b) Decreases.

(c) Doesn't change.

(d) Vanishes.

(ii) switch K_2 only?

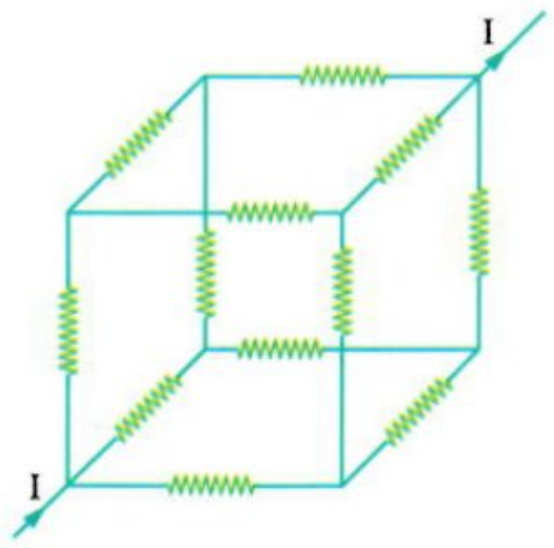
(a) Increases.

(b) Decreases.

(c) Doesn't change.

(d) Vanishes.

52 ✳ In the shown figure, 12 resistors each of resistance R, are connected together in the form of a cube, so the equivalent resistance of the shape in terms of R equals ..... .



(a) $\frac{1}{2}$ R

(b) $\frac{5}{6}$ R

(c) $\frac{3}{2}$ R

(d) R

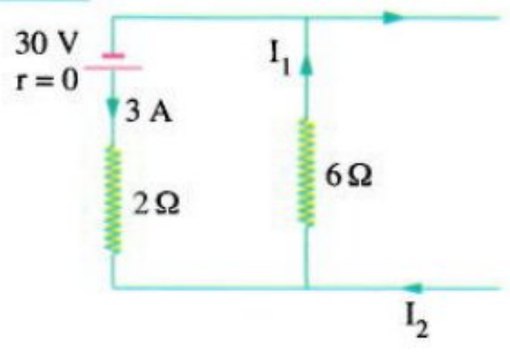
Second Essay questions

1 **Mention the scientific principle on which the following is based:**

(a) Kirchhoff's first law.

(b) Kirchhoff's second law.

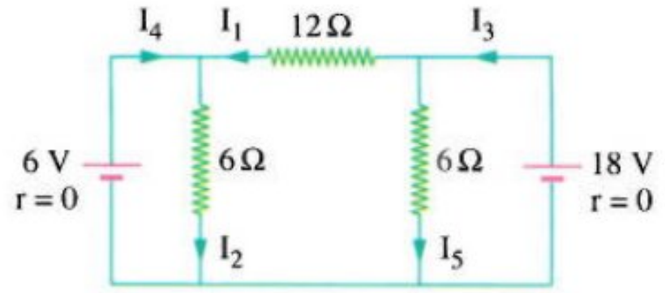
2 The opposite figure represents a part of a closed electric circuit, **calculate** the current intensity I_2



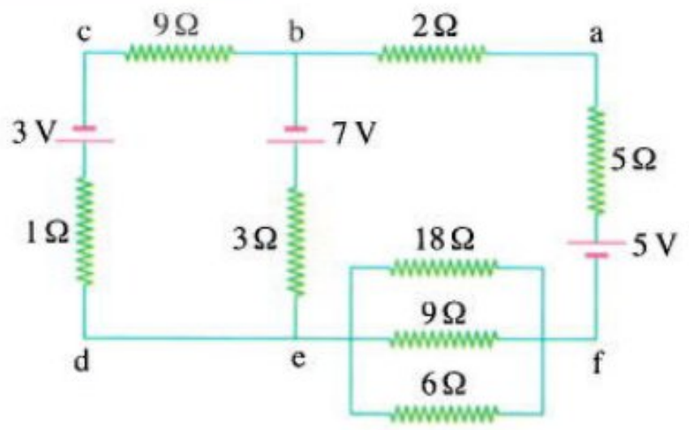
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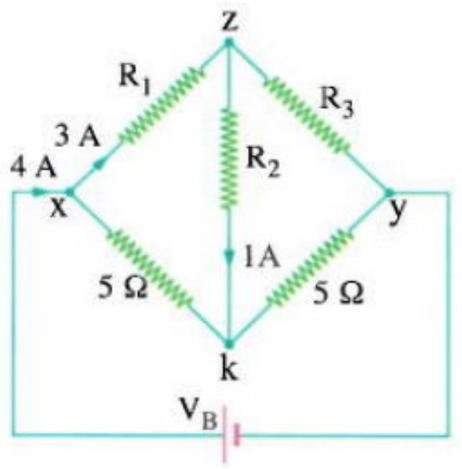
3 **Calculate** current intensities I_3, I_4 in the opposite electric circuit.



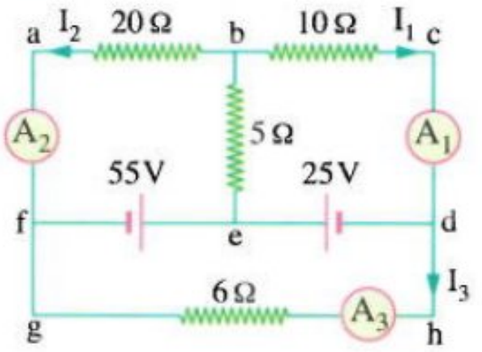
4 From the opposite circuit, **calculate** the current intensity passing through the resistor 3 Ω.



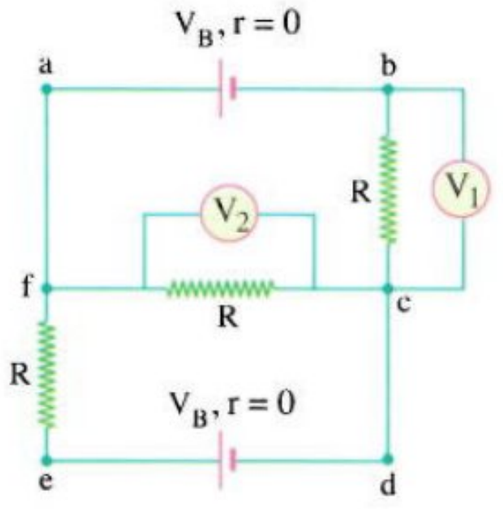
5 The opposite figure represents an electric circuit, **calculate** the emf (V_B) of the battery with neglecting its internal resistance.



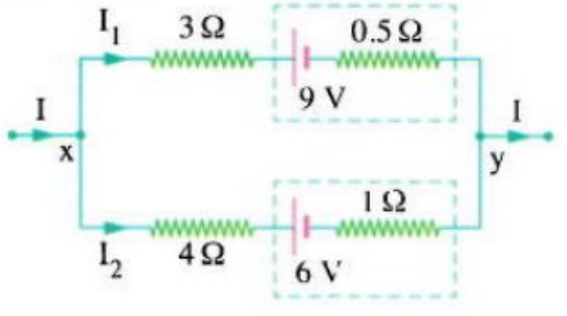
6 In the opposite electric circuit, the reading of ammeter A_1 is 1 A and the internal resistances of the batteries are negligible, **calculate** the ratio between the readings of ammeters A_2 and A_3 $\left(\frac{A_2}{A_3}\right)$.



7 In the opposite electric circuit,
prove that: $\frac{V_1}{V_2} = \frac{1}{2}$



8 The opposite figure represents a part of an electric circuit, if the potential difference between the points x and y equals 16 V, **calculate** the current intensity I.

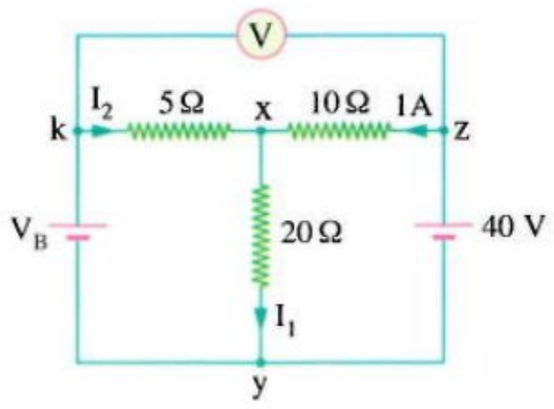


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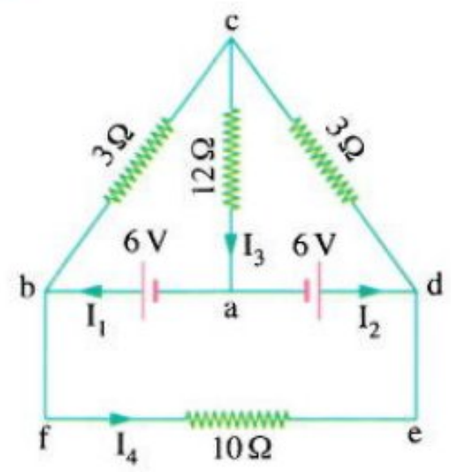
9 In the opposite electric circuit, if the internal resistance of the batteries is negligible, **calculate :**

(a) The voltmeter reading (V).

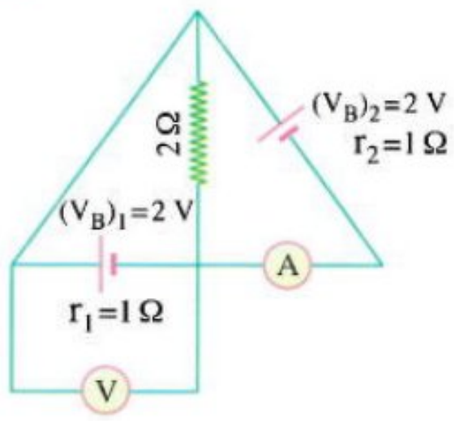
(b) The emf (V_B).



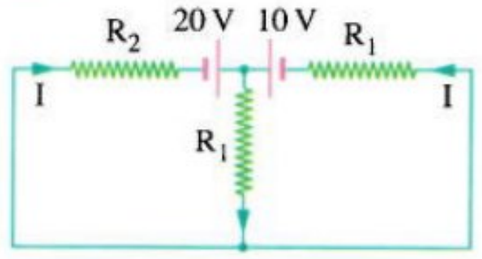
10 From the opposite circuit, **calculate** the current intensity passing in the 12 Ω resistor.



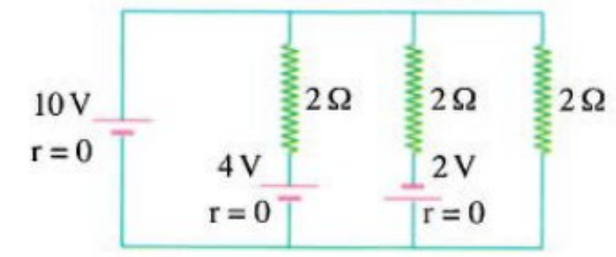
11 In the opposite electric circuit, **find** the reading of each of the ammeter and voltmeter.



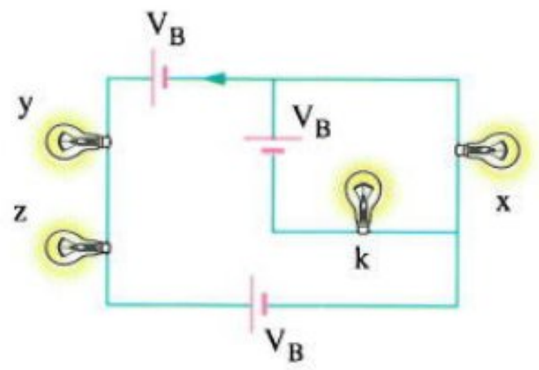
12 From the opposite electric circuit, if the current intensity is the same in each of the two batteries and equals (I) ampere, **find** the ratio between the two resistances $\left(\frac{R_1}{R_2}\right)$.



13 In the opposite electric circuit, **calculate** the current intensity passing in the 10 V battery.



14 The opposite figure illustrates an electric circuit that contains identical batteries of negligible internal resistances and also contains multiple identical bulbs, **which** of the bulbs' filaments glows more intensely?



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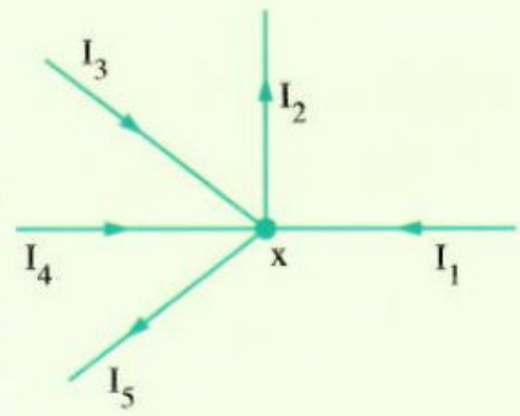
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Questions of the Previous Years Exams ? on Lesson 4

Kirchhoff's first law

1 By applying Kirchhoff's first law at the point (x),

so ..... . *(Exp. June 21)*



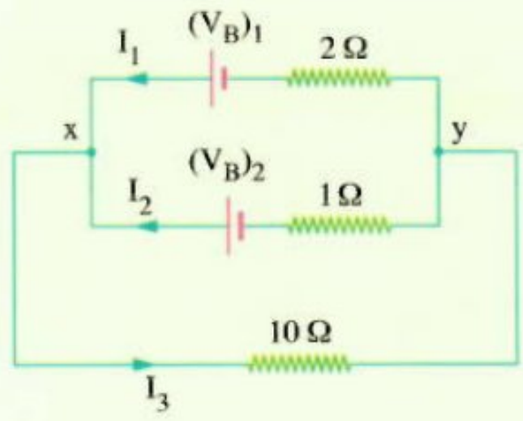
(a) $I_1 + I_3 + I_4 + I_2 + I_5 = 0$

(b) $-I_1 - I_3 - I_4 + I_2 + I_5 = 0$

(c) $-I_1 - I_3 + I_4 + I_2 + I_5 = 0$

(d) $I_1 + I_3 + I_4 - I_2 + I_5 = 0$

2 From the opposite electric circuit, the correct equation is ..... *(2nd session 21)*



(a) $-I_1 - I_2 + I_3 = 0$

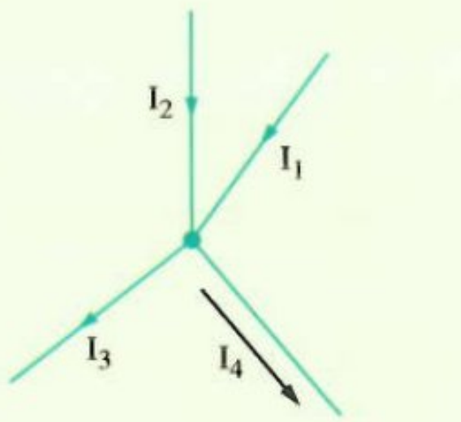
(b) $I_1 - I_2 - I_3 = 0$

(c) $-I_1 + I_2 + I_3 = 0$

(d) $I_1 + I_2 + I_3 = 0$

3 The figure shows a part of a closed electric circuit:

If the directions of I_1, I_2 and I_3 represent the conventional currents while I_4 represents the direction of the free electrons motion. So, current (I_3) = ... *(2nd session 22)*



(a) $I_1 + I_2 - I_4$

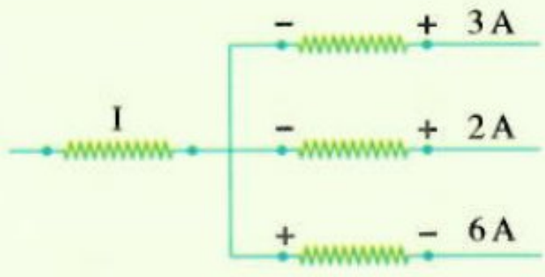
(b) $I_1 + I_2 + I_4$

(c) $I_4 + I_1 - I_2$

(d) $I_4 + I_2 - I_1$

4 The figure shows a part of a closed electric circuit,

then the value of (I) = ... *(2nd session 23)*



(a) 11 A

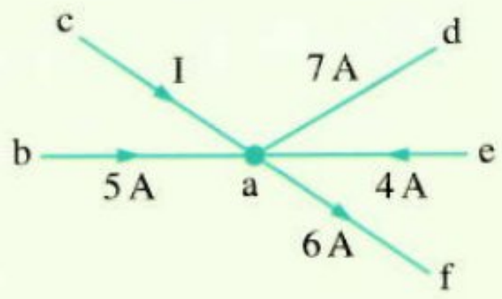
(b) 2 A

(c) 1 A

(d) 4 A

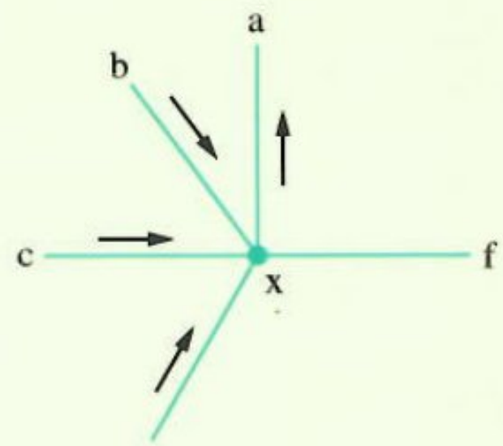
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5 The opposite figure represents electric currents through multiple branches. So, the value of current (I) and the direction of current (7 A) respectively are ..... , ... *(Practice 25)*



(a) 4 A, from a to d
(b) 4 A, from d to a
(c) 13 A, from a to d
(d) 13 A, from d to a

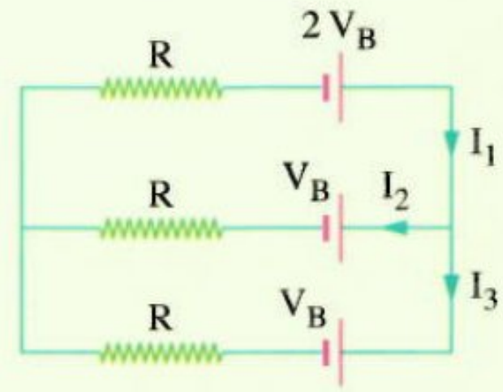
6 In the figure, if the directions shown represent the directions of electrons motion, and the intensities of the currents in branches xa, xc and xd are 6 A, 3 A, 15 A respectively, and 2.5×10^{19} electrons/second pass through branch xb, then the number of electrons passing through branch xf and their direction of motion are respectively ..... , ... *(Practice 25)*



(a) 1×10^{20} electrons/second, from x to f
(b) 1×10^{20} electrons/second, from f to x
(c) 1.6×10^{19} electrons/second, from x to f
(d) 1.6×10^{19} electrons/second, from f to x

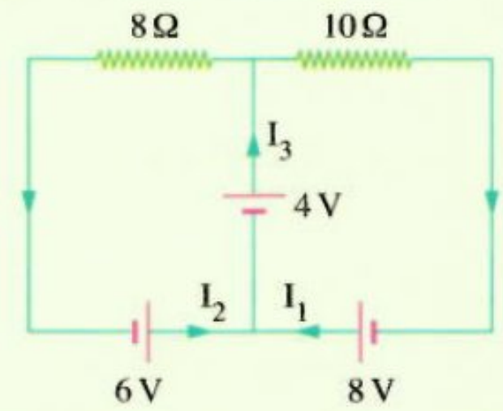
Kirchhoff's second law

7 Using the data shown on the opposite circuit, the ratio $\left(\frac{I_2}{I_1}\right)$ = ... *(Exp. May 21)*



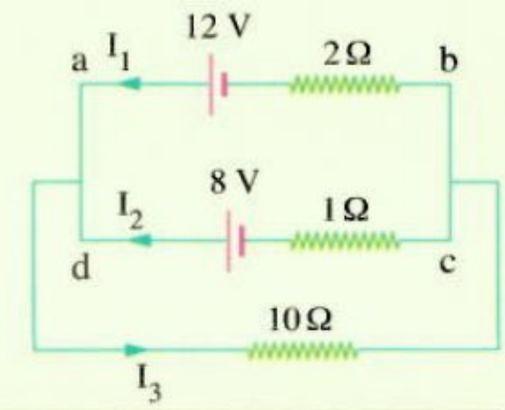
(a) $\frac{2}{1}$
(b) $\frac{3}{1}$
(c) $\frac{1}{2}$
(d) $\frac{1}{3}$

8 In the opposite circuit, the value of electric current I_3 is ..... . *(Exp. June 21)*



(a) 1.2 A
(b) 1.25 A
(c) 2 A
(d) 2.45 A

9 In the opposite electric circuit, we can apply Kirchhoff's laws to the closed path (adcba) as follows ..... . *(1st session 21)*



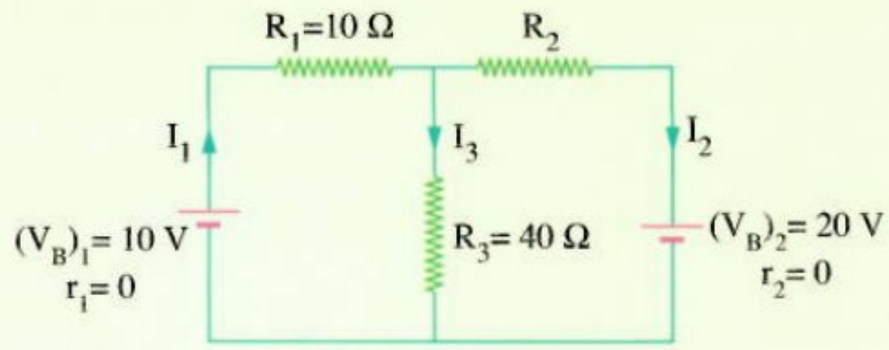
(a) $3 I_1 - I_3 - 4 = 0$
(b) $2 I_1 + I_2 + 4 = 0$
(c) $2 I_1 - I_2 - 20 = 0$
(d) $2 I_1 - I_2 + 4 = 0$

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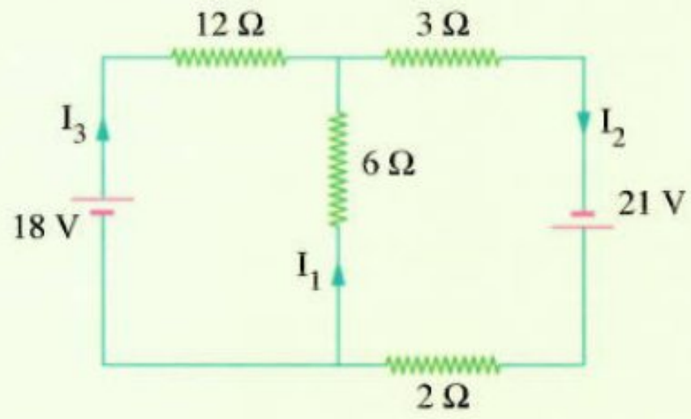
10 In the opposite circuit, if $(I_3 = -2\,I_1)$, the value of the passing electric current in resistor R_3 equals ..... . *(1st session 21)*



(a) $\frac{3}{7}$ A (b) $\frac{4}{7}$ A

(c) 1 A (d) $\frac{2}{7}$ A

11 In the shown electric circuit, if $I_3 = 2$ A, the value of I_2 is ..... . *(2nd session 21)*



(a) 1 A

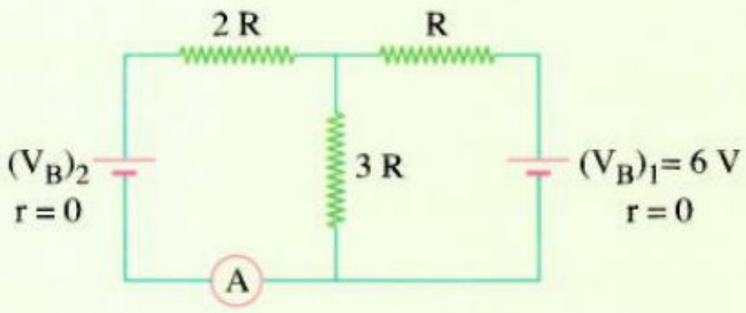
(b) 2 A

(c) 3 A

(d) 4 A

12 In the shown electric circuit:

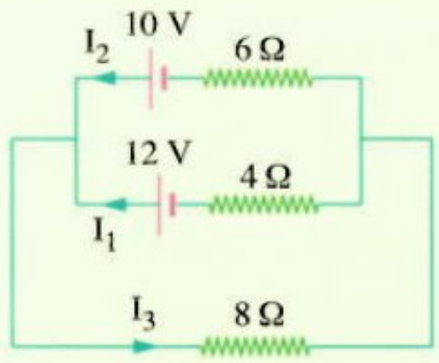
The value of $(V_B)_2$ that makes the reading of the ammeter vanish = ... *(1st session 22)*



(a) 6 V (b) 4.5 V

(c) 8 V (d) 12 V

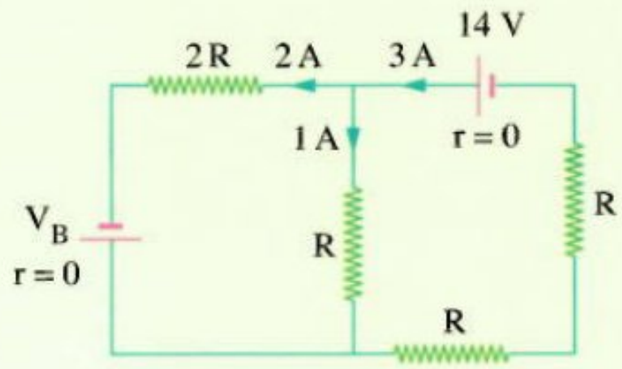
13 In the shown electric circuit, the value of the intensity of electric current passing through the 8 Ω resistor equals ..... *(1st session 22)*



(a) 0.23 A (b) 0.846 A

(c) 1.077 A (d) 1.306 A

14 In the shown closed electric circuit, the value of V_B = ... *(2nd session 22)*



(a) 10 V

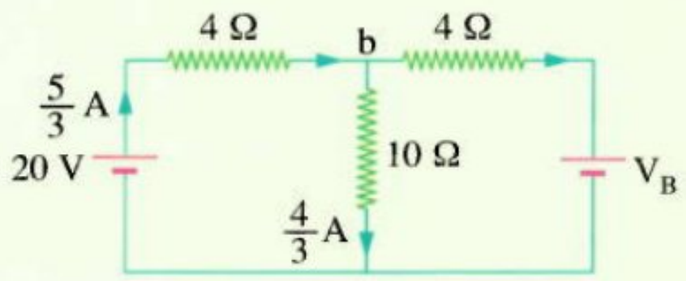
(b) 4 V

(c) 15 V

(d) 6 V

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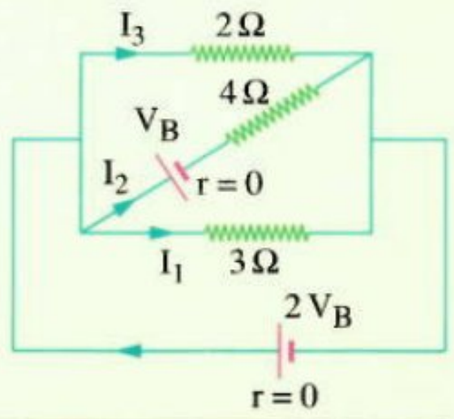
15 In the illustrated electric circuit, the electromotive force of the battery V_B is ..... . *(Exp. 23)*



(a) 12 V (b) $\frac{4}{3}$ V

(c) $\frac{40}{3}$ V (d) $\frac{44}{3}$ V

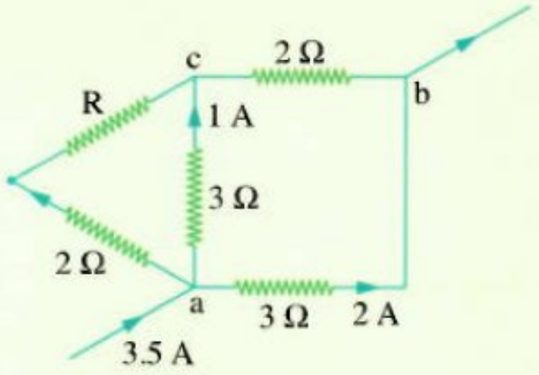
16 In the shown closed electric circuit, the ratio between: $\frac{I_3}{I_2}$ = ... *(1st session 23)*



(a) $\frac{2}{1}$ (b) $\frac{1}{4}$

(c) $\frac{1}{2}$ (d) $\frac{4}{1}$

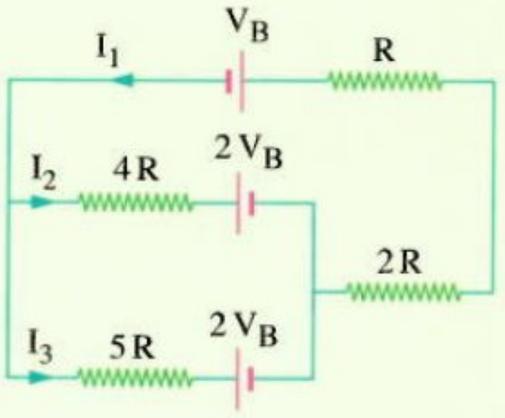
17 The figure illustrates a part of a closed electric circuit. Then the resistance (R) equals ..... . *(2nd session 23)*



(a) 4 Ω (b) 2 Ω

(c) 6 Ω (d) 3 Ω

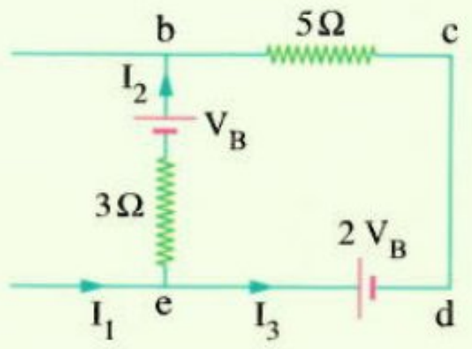
18 From the shown electric circuit, I_1 = ... I_3 *(1st session 24)*



(a) 2.25 (b) 1.25

(c) 0.8 (d) 1.8

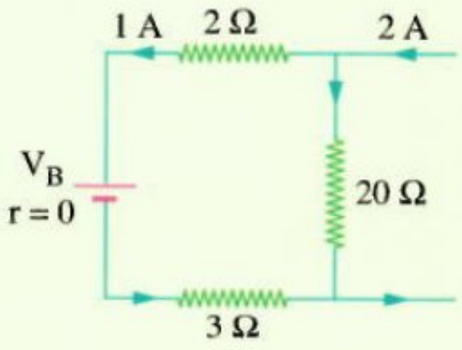
19 The opposite figure shows a part from an electric circuit, by using Kirchhoff's laws, which equation is correct? *(2nd session 24)*



(a) $3 I_1 + 7 I_2 = -3 V_B$ (b) $3 I_2 - 5 I_3 = -3 V_B$

(c) $3 I_2 - 5 I_3 = 3 V_B$ (d) $3 I_1 - 8 I_2 = 3 V_B$

20 The opposite figure represents a section of electric circuit that carries an electric current, then the value of V_B is ..... . *(Practice 25)*



(a) 30 V (b) 25 V

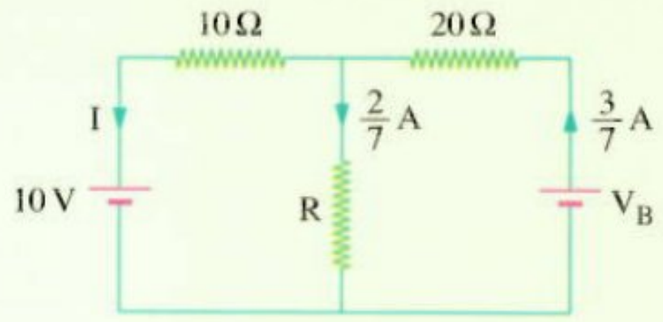
(c) 20 V (d) 15 V

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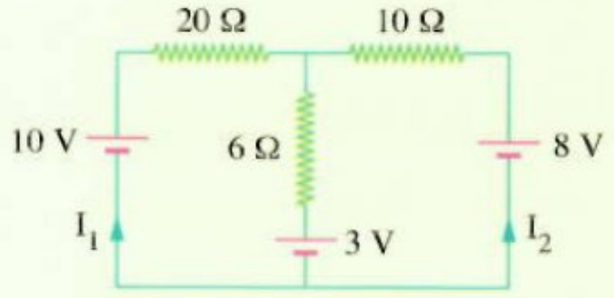
21 In the electrical circuit shown in the figure, what is the value of each of (V_B) and (R)?



	Value of (V_B)	Value of (R)
a	30 V	40 Ω
b	20 V	30 Ω
c	30 V	20 Ω
d	20 V	40 Ω

(1st session 25)

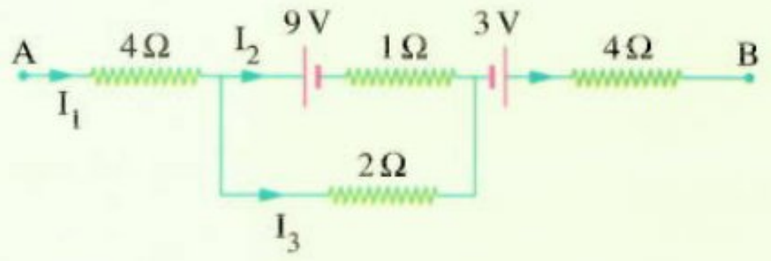
22 In the electrical circuit shown in the figure, what is the value of (I_1) and (I_2)?



	Value of (I_1)	Value of (I_2)
a	0.215 A	0.447 A
b	0.215 A	0.232 A
c	0.232 A	0.132 A
d	0.232 A	0.215 A

(2nd session 25)

23 The opposite figure represents part of an electric circuit. If the potential difference between points A and B is 16 V, then the intensity of the current passing through the 2 Ω resistor equals ..... . *(Practice 25)*



(a) 2.5 A (b) 3.5 A

(c) 4 A (d) zero

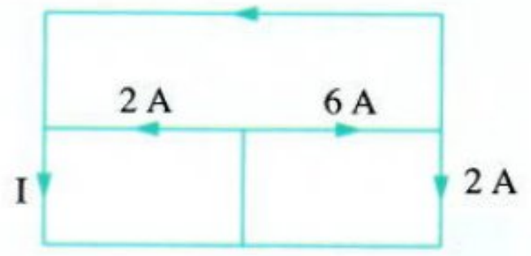
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Test on Chapter 1

Answered

First Choose the correct answer

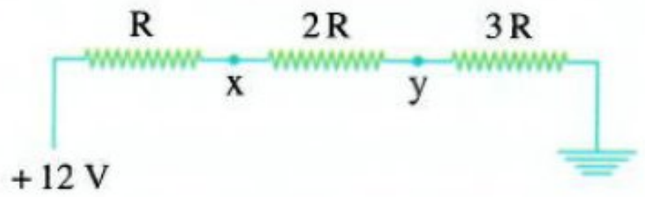
1 In the opposite figure, the value of I is ..... .



(a) 8 A (b) 6 A

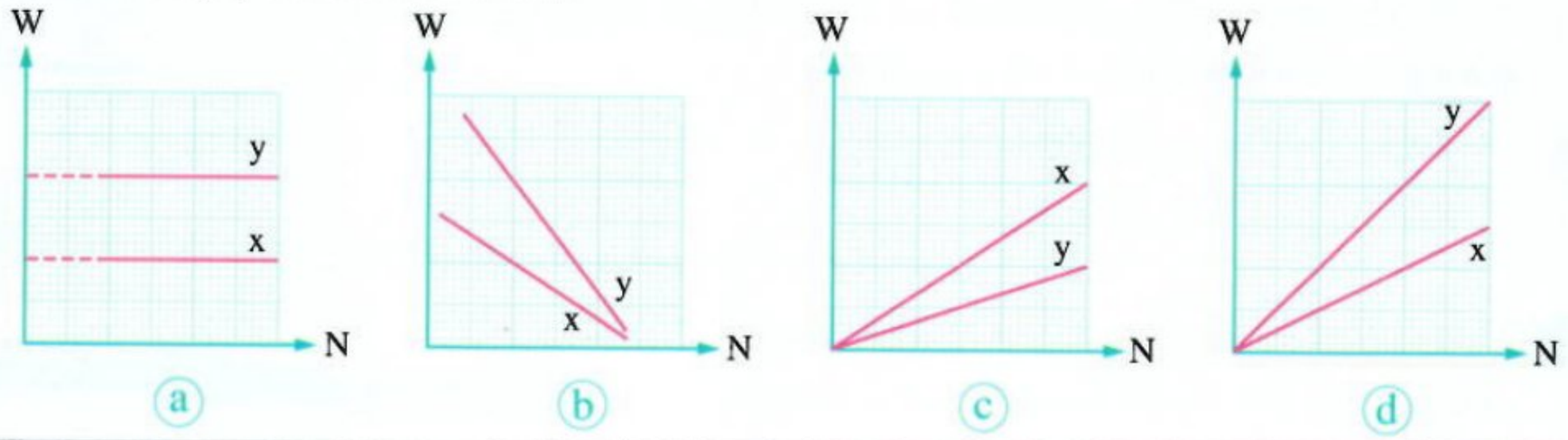
(c) 4 A (d) 3 A

2 The opposite figure shows a section of electric circuit, so the ratio between the potentials of the two points x and y shown on the figure $\left(\frac{V_x}{V_y}\right)$ equals ..... .

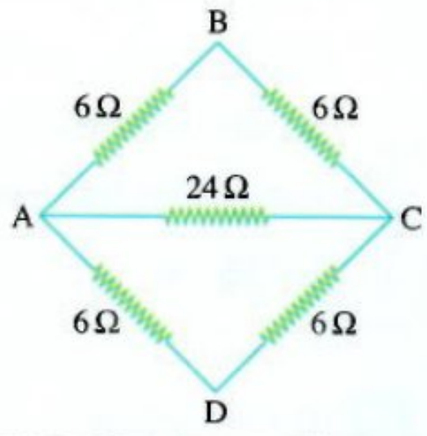


(a) $\frac{3}{5}$ (b) $\frac{5}{3}$ (c) $\frac{1}{3}$ (d) $\frac{3}{1}$

3 Two conductors x and y have electrical resistances R and 2 R respectively, each of them carries an electric current of intensity I, which of the following graphs represents the relation between the work done (W) to transfer electrons across the two conductors and the number (N) of these electrons?



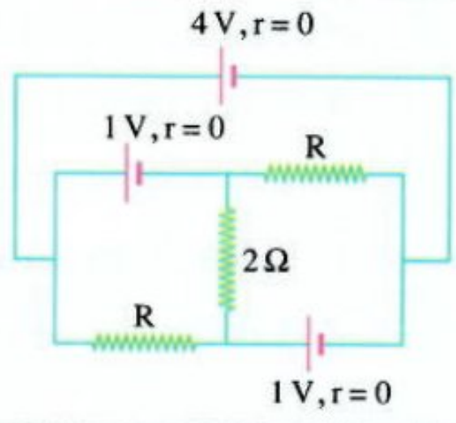
4 A set of electric resistors are connected together as shown in the opposite figure, so this set has a higher equivalent resistance when connecting a battery to the two terminals ..... .



(a) A, C (b) A, B

(c) B, D (d) C, D

5 The electric current intensity through the 2 Ω resistor in the opposite electric circuit equals ..... .



(a) 0.5 A (b) 1 A

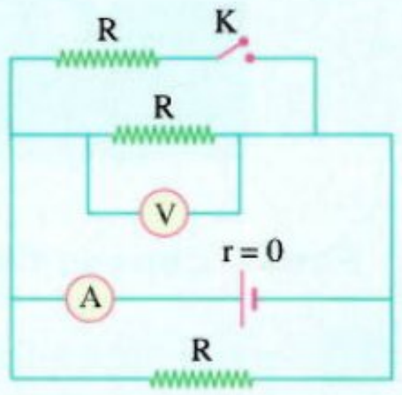
(c) 2 A (d) 4 A

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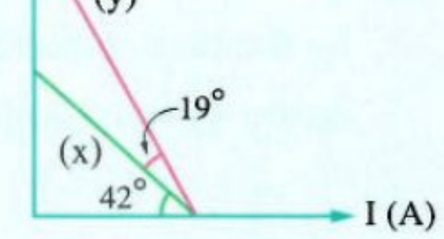
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6 What happens to the readings of both ammeter and voltmeter respectively when closing switch K in the opposite electric circuit?



(a) Increases, doesn't change.
(b) Doesn't change, doesn't change.
(c) Increases, increases.
(d) Doesn't change, increases.

7 Each of two electric cells (x) and (y) is connected to a different electric circuit. The opposite figure represents the graph of voltages (V) across the two electric cells versus current intensity (I) through each of them, so the ratio between the two electromotive forces of the electric cells $\left(\frac{(V_B)_x}{(V_B)_y}\right)$ is approximately equal to ..... .



(a) 0.2 (b) 0.3 (c) 0.4 (d) 0.5

8 Each of the two opposite figures (1) and (2) is a section of an electric circuit through which a direct current is passing, so the electric energy consumed by resistance R in figure (2) through a time interval of 10 s equals ..... .

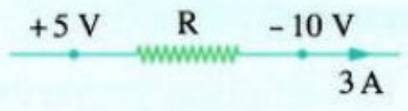


Figure (1)

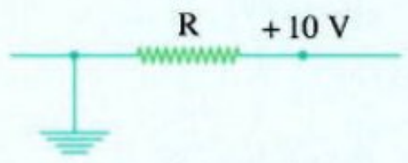
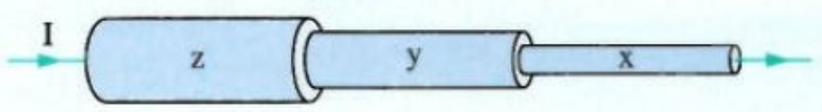


Figure (2)

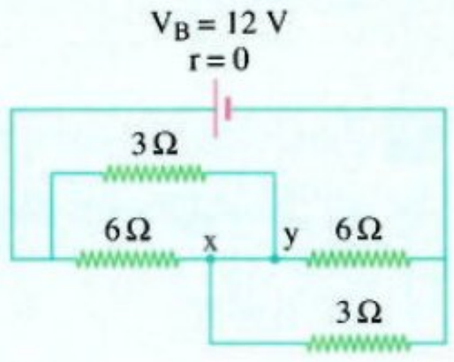
(a) 10 J (b) 20 J
(c) 100 J (d) 200 J

9 Three solid conductors x, y and z are made of the same metal and have the same length, their cross-sectional diameters are d, 2d, 3d respectively, so if they are connected together in series, then a direct electric current is set up through them as shown in the opposite figure, so the ratio of potential differences across the three conductors $V_x : V_y : V_z$ respectively are ..... , ...



(a) 36 : 9 : 4 (b) 9 : 4 : 1 (c) 3 : 2 : 1 (d) 6 : 3 : 2

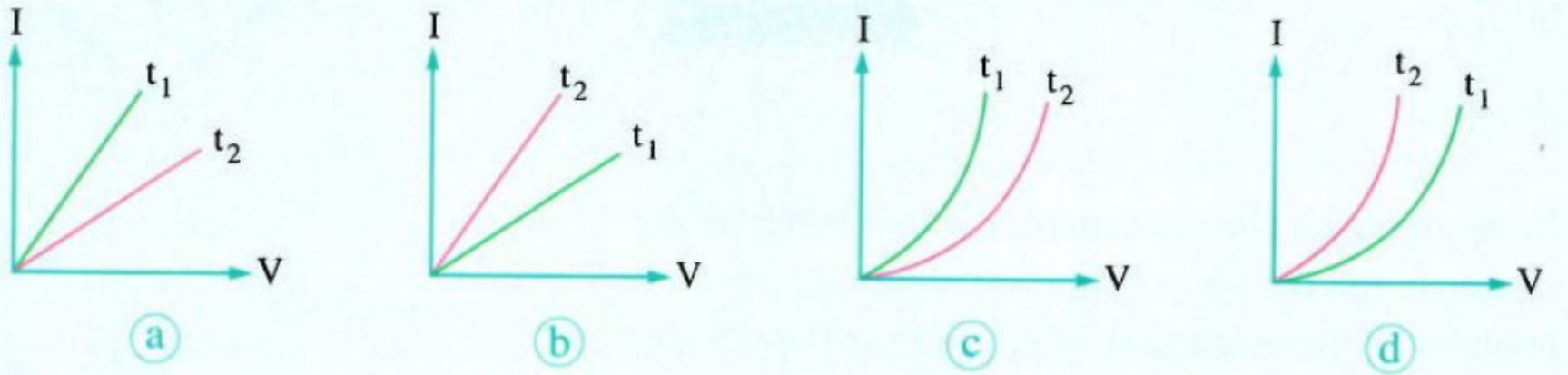
10 In the opposite electric circuit, the intensity and the direction of electric current between points x and y are ..... .



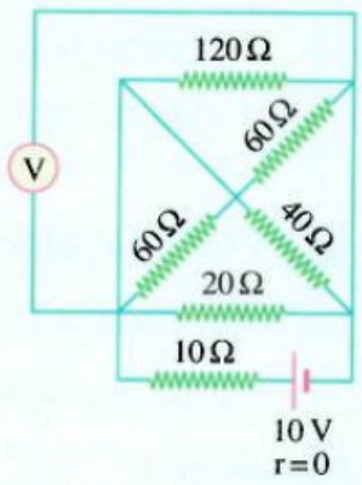
(a) 0.5 A from y to x
(b) 1 A from y to x
(c) 0.5 A from x to y
(d) 1 A from x to y

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11 An Ohm's law experiment has been conducted for determining the resistance of a metal wire at a constant temperature t_1 then after raising the temperature of the wire to t_2. So, which of the following graphs could represent the experiment results?

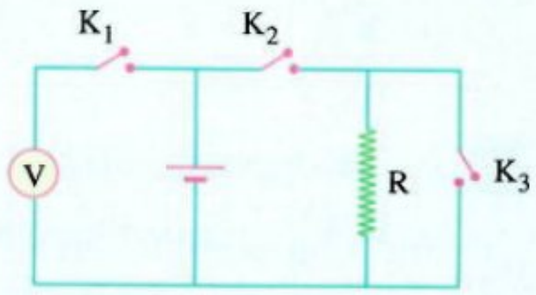


12 In the opposite electric circuit, the reading of the voltmeter (V) is ..... .



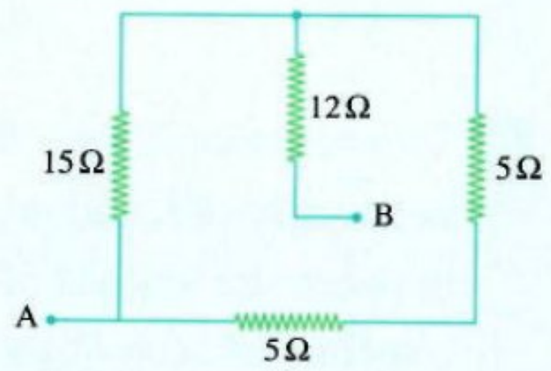
(a) 10 V
(b) 5 V
(c) 2.5 V
(d) 1 V

13 In the opposite electric circuit, a voltmeter is connected across a battery of emf V_B and internal resistance r, so the voltmeter shows the highest possible reading in this circuit when closing ..... .



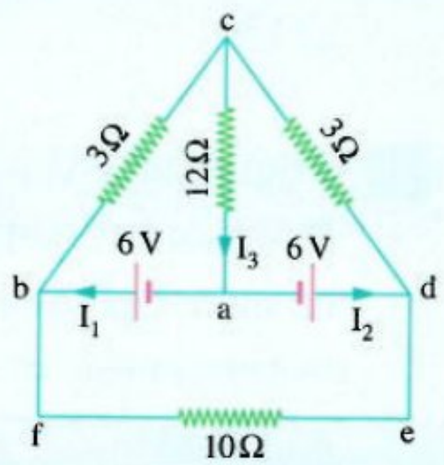
(a) switches K_1 and K_3 only
(b) switches K_1 and K_2 only
(c) switches K_2 and K_3 only
(d) all switches K_1, K_2 and K_3

14 In the opposite figure, when connecting a battery between points A and B, an electric current of intensity 2 A passes through the 12 Ω resistor, so potential difference across the battery terminals equals ..... .



(a) 10 V
(b) 12 V
(c) 20 V
(d) 36 V

15 In the opposite electric circuit, the intensity of current through the 12 Ω resistor equals ..... .

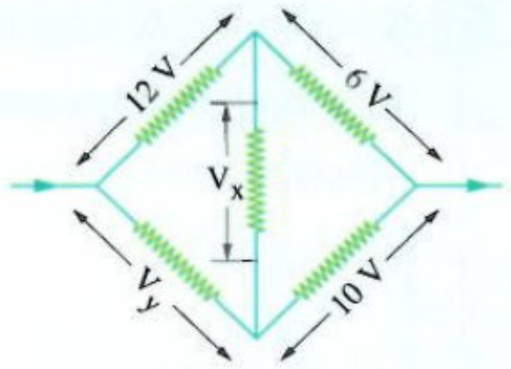


(a) 0.11 A
(b) 0.22 A
(c) 0.33 A
(d) 0.44 A

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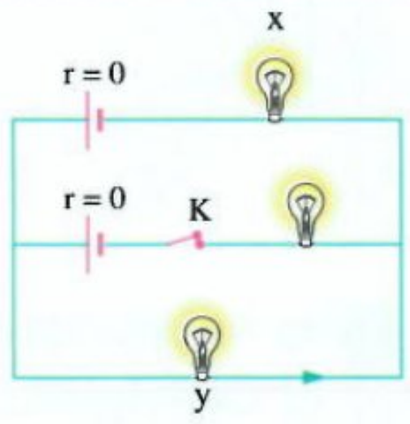
16 The opposite figure illustrates a section of electric circuit, so the ratio of voltages $\left(\frac{V_x}{V_y}\right)$ equals ..... .



(a) $\frac{1}{1}$ (b) $\frac{1}{2}$

(c) $\frac{2}{1}$ (d) $\frac{1}{4}$

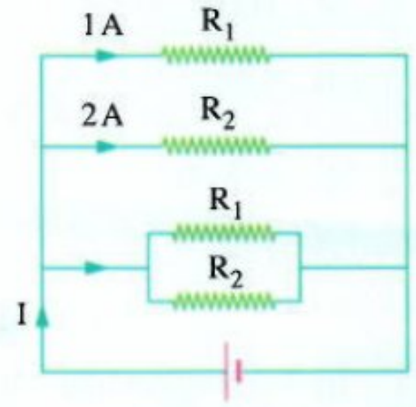
17 In the opposite electric circuit, the two batteries are identical and the three lamps are identical, what happens to the brightness of each of the two lamps x, y respectively when opening switch K?



(a) Increases, increases. (b) Decreases, decreases.

(c) Increases, decreases. (d) Decreases, increases.

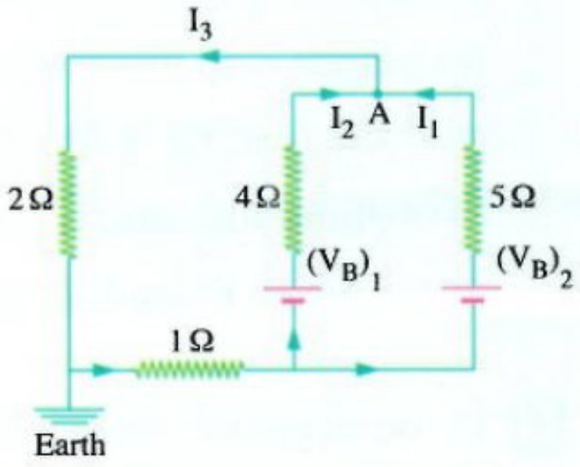
18 The total electric current intensity (I) that passes in the opposite electric circuit equals ..... .



(a) 5 A (b) 6 A

(c) 8 A (d) 9 A

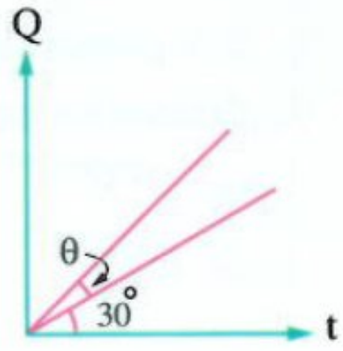
19 In the opposite electric circuit, the two batteries $(V_B)_1$ and $(V_B)_2$ have electromotive forces 7 V and 10.5 V respectively and the internal resistance of each of them is 1 Ω, the electric potential of point A equals ..... .



(a) 1 V (b) 2 V

(c) 3 V (d) 5 V

20 Two resistors R and 2R are connected together in parallel with an electric cell, and the opposite graph represents the relation between the amount of electric charge (Q) that passes through a section of each of them and the time (t), so angle θ which is shown in the graph equals ..... .



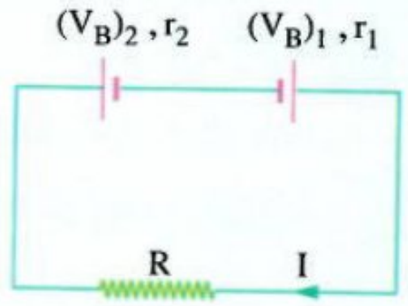
(a) 15° (b) 19.1° (c) 30° (d) 49.1°

21 A lightbulb (A) is labelled with (240 V, 480 W), a lightbulb (B) is labelled with (140 V, 70 W) and a lightbulb (C) is labelled with (100 V, 50 W), so the equivalent resistance of the three lightbulbs after connecting them together such that each of them works with its maximum power equals ..... .

(a) 236 Ω (b) 284 Ω (c) 149 Ω (d) 96 Ω

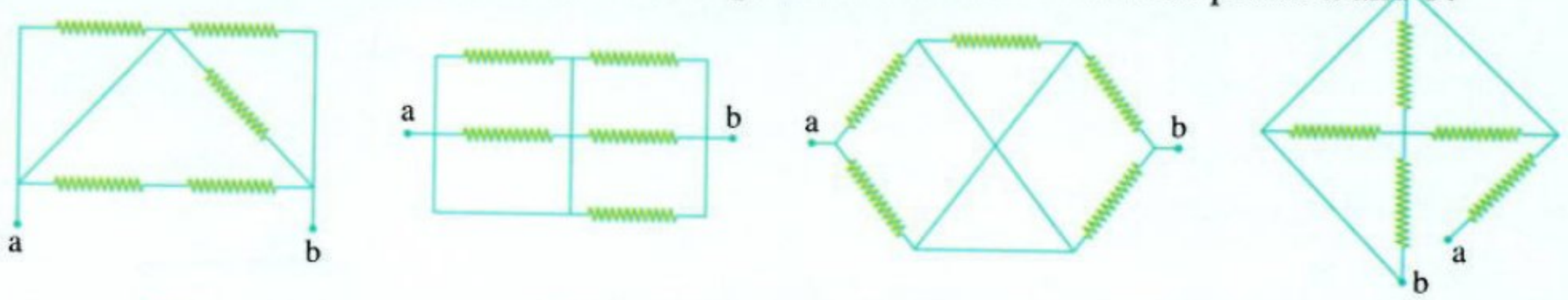
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22 In the opposite electric circuit, an electric current of intensity I is passing, if when reversing the terminals of battery $(V_B)_1$, the current intensity through the circuit increases to 3I, so the ratio $\left(\frac{(V_B)_1}{(V_B)_2}\right)$ equals ..... .



(a) $\frac{2}{1}$ (b) $\frac{3}{1}$ (c) $\frac{4}{1}$ (d) $\frac{5}{1}$

23 The following figures illustrate four methods of connecting five resistors, each of resistance R, so which of these methods gives the least equivalent resistance between points a and b?



(a) (b) (c) (d)

24 Two resistance wires x and y are made of the same material, each has a uniform cross-section and the length of wire x is double that of wire y. When the two wires were connected in series in an electric circuit, the potential differences across them were V and 2 V respectively, so the ratio between the masses of the two wires $\left(\frac{m_x}{m_y}\right)$ equals ..... .

(a) $\frac{4}{1}$ (b) $\frac{1}{4}$ (c) $\frac{8}{1}$ (d) $\frac{1}{8}$

25 Figure (1) represents the graph of the readings of voltmeters (V_1) and (V_2) versus the ammeter reading (I) in the electric circuit illustrated in figure (2):

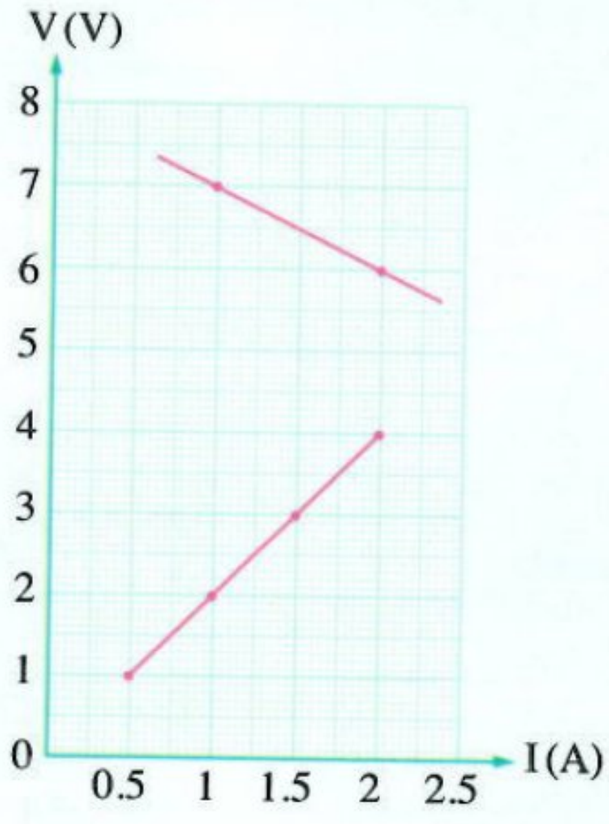


Figure (1)

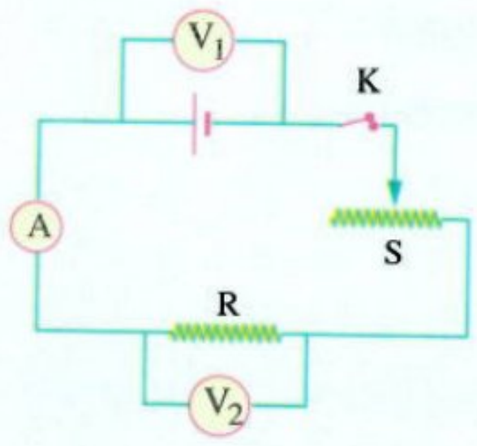


Figure (2)

So, the resistance of S when the ammeter reading is 1.25 A, equals ..... .

(a) 1.7 Ω (b) 2 Ω (c) 3.4 Ω (d) 6.4 Ω

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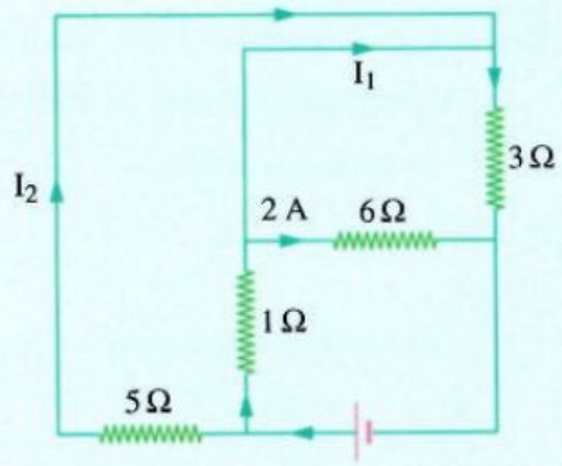
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26 Two wires are made of the same metal, the first has resistance R_o while the ratio between its length to its diameter equals x and the second has resistance nR_o while the ratio between its length to its diameter is y, so the ratio between the radii of the two wires $\left(\frac{r_1}{r_2}\right)$ equals ..... .

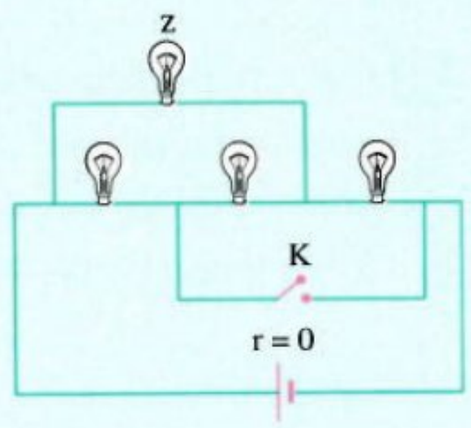
(a) $\frac{x}{y}$ (b) $\frac{y}{x}$ (c) $n\frac{x}{y}$ (d) $n\frac{y}{x}$

27 In the opposite electric circuit, the intensities of currents I_1 and I_2 respectively are ....., ...



(a) 4 A, 1 A
(b) 2 A, 2 A
(c) 3 A, 5 A
(d) 3 A, 1 A

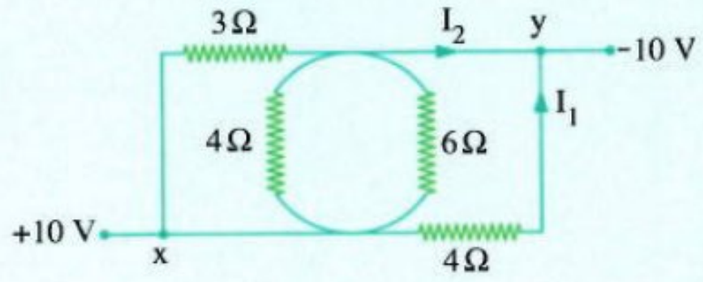
28 Four identical electric lightbulbs are connected in an electric circuit as shown in the opposite figure. Given that the internal resistance of the battery is negligible, so when closing switch K, the brightness of lightbulb z ..... .



(a) increases (b) decreases
(c) vanishes (d) doesn't change

Second Answer the following questions

29 The opposite figure illustrates a section of an electric circuit, **calculate:**



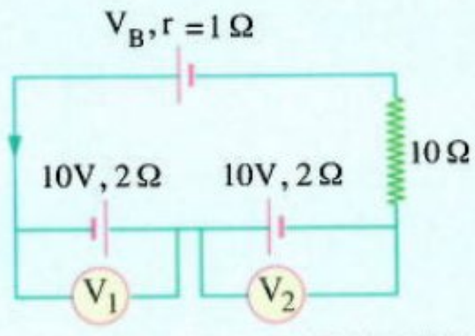
(a) Current intensity I_1

(b) Current intensity I_2

.....

.....

30 In the opposite electric circuit, if the voltmeter reading V_1 equals zero, **find**:



(a) The emf of battery V_B

(b) The reading of voltmeter V_2

.....

.....

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